



A comparison of five reflective–formative estimation approaches: reconsideration and recommendations for tourism research

Jun-Hwa Cheah¹ · Hiram Ting² · T. Ramayah³ · Mumtaz Ali Memon⁴ · Tat-Huei Cham⁵ · Enrico Ciavolino⁶ 

Published online: 29 October 2018
© Springer Nature B.V. 2018

Abstract

In partial least squares structural path modelling, the reflective–formative type of hierarchical component models (HCMs) (also known as Higher-Order Model) have become a popular choice for researchers. However, current approaches to estimate the reflective–formative type of HCM are ambiguous especially when used as an endogenous construct or a mediator. This paper presents a comparison between five different approaches (repeated indicator, two types of two-stage, hybrid, and improved repeated indicator) with two different estimation modes (Mode A and Mode B) when modelling a mediator construct of a reflective–formative HCM in the structural model. By using a model based on stimulus–organism–response theory, an empirical application to the tourism field is adopted in this study. The proposed HCM model examines perceived relative advantages as a mediation of the relationship between Communicability and Intention to Purchase Travel Online. The findings suggest that the improved repeated indicator approach with Mode B estimation yields better path coefficients, goodness of fit, explained variance, and predictive relevance as compared to other approaches. The study provides valuable recommendations and guidelines for tourism researchers to properly conduct an HCM analysis.

Keywords PLS-PM · Hierarchical component model (HCM) · Reflective–formative · Tourism

1 Introduction

The use of partial least squares path modelling (PLS-PM) (Wold 1982; Lohmöller 1989; Hair et al. 2017b) allows researchers to more efficiently examine and substantiate complex theoretical models in a plethora of fields. For instance, PLS-PM has been widely adopted in disciplines such as accounting (Lee et al. 2011; Nitzl 2016), family businesses (Sarstedt et al. 2014), group and organization management (Sosik et al. 2009), hospitality (Ali et al. 2018), human resource management (Ringle et al. 2018), information systems (Ringle

✉ Enrico Ciavolino
enrico.ciavolino@unisalento.it

Extended author information available on the last page of the article

et al. 2012; Hair et al. 2017a), international marketing research (Henseler et al. 2009; Richter et al. 2016a, b), marketing (Hair et al. 2012), operations management (Peng and Lai 2012), psychology (Willaby et al. 2015), strategic management (Hair et al. 2012), supply chain management (Kaufmann and Gaeckler 2015), and tourism (do Valle and Assaker 2016). The variance-based composite PLS-PM method is widely popular among researchers from many disciplines because of its capability to explain and predict the phenomenon of interest. Furthermore, Wold (2006) notes that in large and complex models with latent variables, PLS-SEM is “virtually without competition.” As noted by Richter et al. (2016, p. 590), additional reasons for using PLS-PM are “the use of composites that represent formative latent variables, and the use of small sample sizes due to a small population. By applying PLS-PM latent variable scores in subsequent analyses, it is possible to overcome factor-based limitations of SEM’s by mimicking the results of common factor models” (see also Rigdon 2016; Sarstedt et al. 2016). In summary, “(s)oft modelling is primarily designed for research contexts that are both data-rich and theory-skeletal.” (Wold 1982, p. 29).

Due to the success of PLS-PM, researchers are calling for the more rigorous use of PLS-PM and its results assessment (e.g., Goodhue et al. 2012; Marcoulides et al. 2012; Rönkkö and Evermann 2013; Henseler et al. 2014), especially considering the increasing complexity of theories and cause-effect models used in business research (Hair et al. 2013; Sarstedt et al. 2017). Much current research focuses on how to properly use hierarchical component models, HCMs (Lohmöller 1989), which also referred to as higher-order constructs or second-order models (e.g., Edwards 2001; Jarvis et al. 2003; Wetzels et al. 2009; Becker et al. 2012; Johnson et al. 2012; Polites et al. 2012; Ringle et al. 2012; van Riel et al. 2017). HCM refers to a more general construct that is measured at a higher level of abstraction, while simultaneously assessing several sub-components (dimensions). Hence, by specifying lower-order components, HCM cover concrete traits of a more general conceptual variable of interest (Hair et al. 2018). This type of modelling allows researchers to reduce the number of structural model relationships. This makes the model more parsimonious and increases the content comprising specific constructs (e.g., Becker et al. 2012). In addition, introducing HCM into a structural model can reduce multicollinearity among the lower order constructs, or formative indicators that exhibit high levels of collinearity (e.g., Becker et al. 2012).

When implementing an HCM, researchers have four main types to choose from (Jarvis et al. 2003; Wetzels et al. 2009): (1) reflective-reflective, (2) reflective-formative, (3) formative-reflective, and (4) formative-formative. Each type depicts the specific relationship between higher order constructs (HOC) and lower order constructs (LOCs), as well as the measurement model used to operationalize the constructs on the lower-order level (Hair et al. 2018). In recent years, the reflective-formative HCM has become the most widely used in empirical applications. It is notable that several studies have been published that describe different methods used to estimate reflective-formative HCMs (e.g., Becker et al. 2012; Ciavolino 2012; Ringle et al. 2012; Ciavolino and Nitti 2013; van Riel et al. 2017). However, few researchers provide a concise explanation of how to use an HOC in the PLS-PM as an endogenous construct or as a mediator. Although Becker et al.’s (2012) stimulation study recommends the use of the repeated indicator (Mode B) and two-stage approaches to estimate HCMs, it remains unclear whether these approaches provide accurate estimates when used with this type of model. This can cause confusion and concern for researchers with respect to the use of the repeated indicator (either through Mode A or Mode B model estimation) for a formative specified construct of an HCM as suggested by Becker et al. (2012). It is also not clear whether the hybrid approach (Mode A and Mode

B) represents a suitable alternative to an HCM assessment because it is seldom used in practice (Wilson and Henseler 2007). Given the lack of current knowledge regarding HCM practices, thus the use of a reflective–formative in the PLS-PM is one reason why it continues to receive limited attention in PLS-SEM literature.

Evidently, this situation remains severe in the tourism and hospitality management field. Sophisticated behavioural or perceptual constructs by means of HCM have become crucial in tourism and hospitality (Bernini et al. 2015; González Rodríguez and Sámper 2012). It is considered to be more effective at making predictions, producing valuable results (bandwidth-fidelity trade-off) and contributing to the body of knowledge. However, empirical assessments of the relative efficacy of the HCM (i.e., reflective–formative) are still scarce in the PLS-PM literature on tourism (do Valle and Assaker 2016) and hospitality management (Ali et al. 2018). For instance, do Valle and Assaker's (2016) critical review identifies that out of 44 tourism articles, only five of them (11.40%) reported the use of HCM in their PLS-PM application. Similarly, Ali et al. (2018) raise the same concern regarding hospitality research, stating that only nine (31.03%) out of 29 articles reported the use of HCM. Although the guidelines on HCM assessment in PLS-PM are documented (Becker et al. 2012; Hair et al. 2018; Ringle et al. 2012; van Riel et al. 2017), the data suggests that most tourism and hospitality researchers are not familiar with the use of HCM.

To address the lack of research, this study contributes to tourism literature by comparing and discussing four established methods. Mode A and Mode B were also used for the assessment of HCMs with the HOC as an endogenous variable or a mediator. In addition, this study proposes an improved repeated indicator approach with Mode A and Mode B to assess reflective–formative HCM. This new approach is useful for estimating composites in PLS-PM. The results of the empirical comparison in this study shows that the improved repeated indicator approach with Mode B is able to approximate a stronger explanatory path coefficient result, explained variance, and predictive relevance when compared to other approaches. Furthermore, Ringle et al.'s' (2012) and Becker et al.'s' (2012) two-stage (Mode A and Mode B) approaches yield identical results for both quality criteria and structural assessment. It is suggested that the hybrid approach (Mode A and Mode B) be used with great caution to assess HCM as an endogenous construct or a mediator because they underestimate the correlation results between the HCM and other constructs in the nomological net. Finally, the study provides recommendation on how to run an improved repeated indicator approach as an alternative method for assessing reflective–formative HCM measured constructs in PLS-PM.

2 HCM modelling and estimation approaches in PLS-PM

Hierarchical component models (HCMs) often include a range of multidimensional reflective measurements shown as lower order constructs (LOCs), to form the higher order construct (HOC) (Jarvis et al. 2003). As a result, the type of measurement model used for LOCs and the HOCs needs to be determined separately (Hair et al. 2018) and must have theoretical support (Ciavolino et al. 2018; Nitti and Ciavolino 2014). A widely debated HCM configuration is the reflective-reflective type. Critics argue that this type of HCM does not exist, or that it is meaningless because the reflective measures should be unidimensional and conceptually interchangeable. This conflicts with the view held by others that multiple underlying dimensions are distinct in nature (Cadogan and Lee 2013; Rigdon 2014a, b). This may explain the popularity of reflective–formative HCM in the

social sciences (Ringle et al. 2012). The LOCs employ a reflective measurement model, whereas the HOC uses a formative measurement model and is formed with data from the LOCs. (Sarstedt et al. 2016). Barroso and Picón (2012) provide a clear example of a reflective–formative HCM in their multidimensional analysis (i.e., benefit loss costs, personal relationship lost costs, economic risk costs, evaluation costs, setup costs, and monetary lost cost) of perceived switching cost (see also Sarstedt et al. 2016). This study builds on an idea from Cadogan and Lee (2013) and looks at the effects of reflective–formative HCM as an endogenous or a mediator on the path between a predictor and an outcome of constructs.

While reflective–formative HCM has received considerable attention, the extant approaches used to estimate such models have been shown to provide inconsistent estimates (Becker et al. 2012). In general, there are three common approaches to estimate HCM in PLS-PM described in the literature. These are (1) the repeated indicator approach (Nitti and Ciavolino 2014; Wold 1982; Lohmöller 1989), (2) the two-stage approach (Becker et al. 2012; Nitti and Ciavolino 2014; Wetzels et al. 2009; Ringle et al. 2012), and (3) the hybrid approach (Wilson and Henseler 2007). Additionally, a recent approach was proposed by van Riel et al. (2017) to assess HCM as composites of common factors using Mode A consistent (or PLS_c) estimation. This approach is advantageous if the data represents a common factor-based population in LOCs to form a HOC composite. The differences between common factor-based and composite-based data populations are described in Schönemann and Wang 1972; Rigdon 2014a, b; Sarstedt et al. 2016). However, it seems unnecessary to mimic the results of factor-based SEM in the LOCs level (factor indeterminacy issue) when the research goal in PLS-SEM is to predict the HCM construct as either an endogenous variable or a mediator. Instead PLS-SEM should adhere to Rigdon's (2012) call to emancipate the method from the factor-analytic tradition and instead focus on developing the HCM (LOCs and HOC) as a purely composite-based statistical methodology (Rigdon 2014a, b; Sarstedt et al. 2016). Therefore it is necessary to make a comparison between the approaches (i.e., the repeated indicator approach, the two-stage approach, and the hybrid approach) and the assessment of HCM based on composite modelling logic.

In the repeated indicators approach, the manifest indicators of the LOCs are reused for the HOC. For example, when a HOC consists of three LOCs with four manifest variables, each of the 12 variables are reused as indicators for the second-order construct. When using the repeated indicator approach, researchers must determine the measurement mode of the HOC with the repeated indicators. Mode A is the recommended method, although Mode B is often used to provide alternative estimates (see Becker et al. 2012; Ringle et al. 2012). This approach has three major disadvantages. First, it can evoke artificially correlated residuals. Second, this estimation only works effectively when the LOCs have a similar number of indicators. Otherwise, the weights of the first order constructs on the second order construct can result in bias estimation (Becker et al. 2012). Third, if a reflective–formative HCM is used as an endogenous construct, almost all of its variance is explained by its indicators. Consequently, there is no remaining variance to be explained by other potential predictors, and the R^2 value will be 1.0. Researchers may also mistakenly conclude that any predictor pointing at such a type of HOC will be approximately zero in terms of the path coefficient result (Ringle et al. 2012). To manage such model constellations, additional steps are required to specify the relationships between the antecedent construct and the LOCs. Particularly, the HOC is not directly affected by other predictors in the model, but only indirectly through the LOCs (see Becker et al. 2012; Ringle et al. 2012). Therefore the effect of the HOC is viewed as being fully mediated by the LOCs. It is referred to as the total effect analysis of formative relationships between the LOCs and the HOC (Hair

et al. 2018). Modelling total effect through LOCs enables researcher to assess the effects of antecedents on formative HOC via the total effect (i.e., the sum of the effects on the LOCs multiplied by the effect of the LOCs on the HOC) (see Appendix of Table 10, particularly 1a and 1b).

With the two-stage approach, the modelling is particularly useful when the HOC of reflective–formative and formative–formative HCMs is explained by antecedent latent variable(s) that are not found in LOCs in the wider nomological net of the PLS path model (Hair et al. 2018). This approach allows other latent variables like predecessors to explain the variance and can result in the identification of significant path relationships. There are two procedures to perform the two-stage approach. The first procedure was suggested by Ringle et al. (2012) that estimates a repeated indicator approach in the first stage and then uses the first stage of LOCs (serve as manifest variables) in a separate second stage of the HOC (see Appendix of Table 10, particularly 2a and 2b). The second procedure suggested by Becker et al. (2012) that estimates the scores of the LOCs in a first-stage model without the HOC. It then uses these first-stage LOCs scores as manifest variables for the HOC in a second-stage analysis (see Appendix of Table 10, particularly 3, 3a and 3b). Generally, both of these procedures can be applied to all types of HCM and Mode A or Mode B can be used for estimation. Since latent variable scores of the first stage become the indicators of the HOCs' measurement models, the assessment can be based on well-established criteria for reflective and formative measurement model assessment (Hair et al. 2017b). The two-stage approach has the advantage of estimating a more parsimonious model as it does not include LOCs. However, separate estimation of the LOC and HOC is also considered to be as one of the disadvantages of this approach because it does not take into account the entire nomological network (Wilson and Henseler 2007).

Third, the hybrid approach is similar to the repeated indicator approach. Researchers can still use Mode A or Mode B. However, each indicator (manifest variables) can only be used once in a model to avoid artificially correlated residuals (Wilson and Henseler 2007). It splits the indicators of each LOC and uses one half to estimate the LOCs and the other half to estimate the HOC, thereby avoiding the repeated use of indicators in the model (for more details on how to execute hybrid approach, see Appendix of Table 10, particularly 4a and 4b). However, this approach still remains unclear how to proceed in cases where there is an odd number of indicators. Also, it does not guide researcher on which specific indicators need to be assigned to the LOCs or the HOC.

To overcome the identified shortcomings, this study introduces the use of the improved repeated indicator approach to estimate HCMs. This approach is an extension of the conventional repeated indicator approach in both Mode A and Mode B. The assignment of all LOC manifest items to the HOC is similar to the conventional repeated indicator approach. Importantly, it is recommended to pre-specify the reliability of (formative) the second-order composites with the improved repeated indicator approach (Mode A or Mode B) when estimating the HOC (see Appendix of Table 10, particularly 5a and 5b). Since the scores of the HOC (second-order composite) are a linear combination of the scores of the LOCs (reflective), thus utilizing the reliability estimates from the LOCs can increase confidence in the relationship between the measure and outcome (validity). Thus, it can produce a more rigorous statistical estimation of a HCM as compare to others approach.

On these grounds, this study analyzes the effectiveness of these five established approaches with different modes (Mode A and Mode B). This is useful in cases when the HCM becomes an endogenous or a mediator in the PLS-PM—as often encountered in tourism and hospitality management research (see Ali et al. 2018; do Valle and Assaker 2016).

3 Study design and data

In PLS, the algorithm estimation of a measurement model to create a proxy in Mode A is based on correlation weights and is represented by arrows pointing from a construct to its indicators, and in Mode B it is based on regression weights, that are represented by arrows pointing from indicators to their construct (Becker et al. 2013; Hair et al. 2017c). Due to this, the empirical comparison of HCM is split into five types of approach with two different types of estimation modes (Mode A or Mode B): (1) the repeated indicator approach with Mode A or (2) Mode B; (3) the Ringle et al.'s (2012) two-stage approach with Mode A or (4) Mode B; (5) the Becker et al.'s (2012) two-stage approach with Mode A or (6) Mode B; (7) the hybrid approach with Mode A or (8) Mode B, and (9) the improved repeated indicator approach with Mode A or (10) Mode B.

This study uses a tourism research model proposed by Amaro and Duarte (2015) to compare HCM approaches. The specific latent variables used in this study are the exogenous stimulus 'Communicability' (COMM), the HOC mediator 'Perceived Relative Advantages' (PRA) which has five LOCs (i.e., convenience, time saving, financial, produce variety, and enjoyment), and the endogenous response construct 'Intention to Purchase Travel Online' (IPTO) (Fig. 1). The theoretical underpinnings of this model stem from the stimuli-organism-response (SOR) model (Meharbian and Russell 1974; Holak and Lehmann 1990; Higgins et al. 2007). The S–O–R model assumes that an organism exposed to psychological or environmental stimuli will process these stimuli personally and respond accordingly (Arora 1982; Slama and Tashchian 1987).

A 3-item scale was adapted from the previous work (Morrison et al. 2001; Li and Buhalis 2006; Amaro and Duarte 2015) carried out to measure travellers' perceptions on COMM. PRA towards purchase travel online was measured using a 14-item scale developed by Amaro and Duarte (2015), and includes five dimensions: convenience (three items), time saving (three items), financial (three items), product variety (three items), and enjoyment (two items). A global single item was developed to allow a holistic assessment of the PRA. Finally, traveller's IPTO was assessed based on two items adapted from Grewal et al. (1998); Limayem et al. (2000); Teo and Yeong (2003); Bigné et al. (2010). Table 7 gives an overview of the measurement items of the study. Each item was established on a

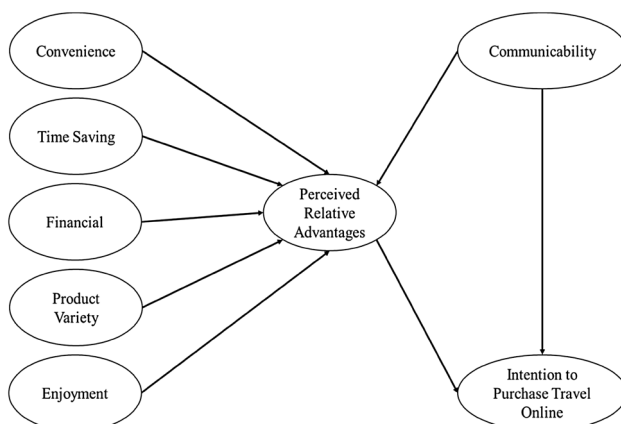


Fig. 1 Empirical research model

seven-point Likert scale, ranging from 1 ('strongly disagree') to 7 ('strongly agree'). The only exception is item INT1, which uses a seven-point Likert scale where 7 represents "Very High" and 1 "Very Low". Questions designed to obtain demographic information like gender, age, ethnicity, education, income, how frequent respondent purchase travel online as well as purpose of purchase travel online by respondent were also included in the questionnaire (see Table 8).

A pre-test allowed for the elicitation of comments and suggestions regarding potential flaws related to the questionnaire format, design, and wording. In addition, it addressed non-response bias and miscomprehension prior to data collection (Fink 2017). Furthermore it assisted to corroborate the measurement of the study through the direction of the causality from construct to measure, which implied by the conceptual definition of the variable (Jarvis et al. 2003). Five consumers from the target population volunteered to participate in the pre-test. Therefore the pilot test was based on the revised questionnaire, and was conducted on an additional 30 respondents so errors could be identified and to optimize the final design of the survey.

Data was collected using an online survey. Using a purposive sampling approach, the questionnaire was distributed among consumers who had purchased travel online on a yearly basis. The data collection yielded 500 usable questionnaires (45 were discarded due to straight lining responses), and it met the minimum sample size of 85 with an effect size of 0.15 and a power level of 80% in the post hoc power analysis (Fink 2017). The respondents were mainly female Malaysians, aged between 21 and 30 years old, who are degree educated, and earn a monthly income ranging from RM2001 to RM4000. The majority of the respondents were annual consumers of travel who possess the intent to go on a family-trip (Table 8 shows a detailed breakdown of the sample characteristics).

A Harman single-factor analysis was performed to ensure that the variance in the data was not explained by one single factor (Podsakoff and Organ 1986). The result indicated that the largest variance explained by the first factor was 30.03%, which is less than 40% of total data variance (Babin et al. 2016; Fuller et al. 2016). Despite collecting data from a single source, common method bias was of no concern and therefore the data did not cause any distortion.

4 Data analysis

In order to corroborate the measurement model in Fig. 1, each type of construct was tested using the confirmatory tetrad analysis (CTA) to determine whether it should be measured in a reflective or formative way (Gudergan et al. 2008; Hair et al. 2018). Subsequently, statistical software SmartPLS 3.2.7 was used to perform the statistical analysis (Ringle et al. 2015). According to Gudergan et al. (2008), the use of CTA distinguishes between formative and reflective measurement models (see Hair et al. 2018; Chapter 3). This helps to avoid the misspecification of measurement models due to incorrect assessments of the HCM in PLS-PM (Jarvis et al. 2003; Diamantopoulos et al. 2008). However, CTA should be used cautiously, especially in studies with a large sample size study. Large sample sizes can lead to a low standard error from the tetrad value estimation, as a result, it can easily lead to rejection of the reflective model. As a result, it is recommended to follow a simple exploratory approach to examine CTA, by running it with a moderate random sample (e.g., $n=200$) of the original data (Bauldry and Bollen 2016; Garson 2016). To adopt this suggestion, 40% of the total sample size was hold-out randomly for the assessment of CTA

(Hair et al. 2010). Additionally, because CTA does not run for latent variables with less than 4 items, this study borrows the indicators of exogenous and endogenous constructs with the highest correlation and with the highest loading (as suggested by Gudergan et al. 2008).

Based on the result of CTA (see Table 9), all tetrad values of the LOCs (i.e., Convenience, Time Saving, Financial, Product Variety, and Enjoyment) vanished in the assessment (Bonferroni-adjusted confidence interval includes zero and insignificant) (Gudergan et al. 2008; Hair et al. 2018). As for the PRA, the result reveals that several tetrads failed to vanish, therefore we can conclude that the construct is not reflective, but a formative HOC (Gudergan et al. 2008; Hair et al. 2018). Based on these findings, the PRA construct is regarded as the type II model of HCM (Jarvis et al. 2003; Becker et al. 2012; Ciavolino and Nitti 2013; Lee and Cadogan 2013), in which the HOC (PRA) is specified as a composite formative measurement, while the LOCs (i.e., Convenience, Time Saving, Financial, Product Variety, and Enjoyment) are specified as reflective measurements (Amaro and Duarte 2015). Dropping one of the LOCs (dimensions) would alter the conceptual domain of PRA. LOCs are not interchangeable in the HCM and any changes in the LOCs would cause changes in PRA's theoretical and conceptual consideration outlined in Amaro and Duarte's (2015) work. Contrastingly, the results of COMM, and IPTO reveal that all tetrads vanished (Gudergan et al. 2008; Hair et al. 2018). Therefore, it can be concluded that the construct of COMM and IPTO is a reflective measurement model because the indicators are the manifestations of the constructs with similar content, and the removal of any indicator does not alter the conceptual domain of the construct.

Drawing from the CTA result, the path model were established and estimated using the repeated indicator approach, both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approaches, the hybrid approach, and the improved repeated indicator approach. All approaches were carried out with indicator weighting of Mode A and Mode B (see Table 10). Before assessing the HCM of the formative measure (PRA), the reliability and validity of the reflective exogenous (COMM) and endogenous (IPTO) constructs was examined. As well as those of the reflective LOCs (i.e., Convenience, Time Saving, Financial, Product Variety, and Enjoyment) (see Hair et al. 2017b, Chapter 4). The assessment of these constructs show that their loadings exceeded the threshold value of 0.7 across all HCM approaches (see Table 1). Likewise, their average variance extracted (AVE) is higher than the critical value of 0.5 and their reliability (i.e., Cronbach's Alpha, Reliability Coefficient ρ_A , and Composite Reliability ρ_C) values are above 0.7. These outcomes substantiate that the reflective measurement of COMM, IPTO, and the LOCs provides valid and reliable results. In addition, since there is only one indicator to measure enjoyment when using the hybrid approach (Mode A and Mode B), it is not possible to estimate internal consistency reliability as well as AVE of the first order construct (see Table 1).

Discriminant validity assessment draws from Henseler, Ringle and Sarstedt (2015) heterotrait–monotrait ratio of correlations (HTMT) measure. The results in Table 2 shows that the HTMT values for the all approaches yield similar results are below the conservative threshold of 0.85 (Kline 2016; Voorhees et al. 2016; Franke and Sarstedt forthcoming). The bootstrapping procedure was run with 10,000 samples and used the no sign changes option, bias-corrected and accelerated (BCa) bootstrap confidence interval, and two-tailed testing at the 0.05 significance level (Streukens and Leroi-Werelds 2016; Aguirre-Urreta and Rönkkö 2018). The results show that none of the HTMT confidence interval includes the value of 1, suggesting that all the HTMT values are significantly different from 1. Additionally, discriminant validity has been established for the (reflective) exogenous and endogenous constructs as well as the LOCs.

Table 1 Reflective measurement model assessment of loading, reliability and average variance extracted

Con- struct	Item	Loading		Internal consistency reliability						Average variance extracted						
		RI (Mode A)	RI (Mode B)	Two stage	Hybrid (Mode A)	Hybrid (Mode B)	RI (Mode A)	RI (Mode B)	Two- stage	Hybrid (Mode A)	Hybrid (Mode B)	RI (Mode A)	RI (Mode B)	Two- stage	Hybrid (Mode A)	Hybrid (Mode B)
Communi- cability	COMM1	0.851	0.851	0.851	0.851	0.851	CA: 0.828	CA: 0.828	CA: 0.828	CA: 0.828	CA: 0.828	0.744	0.744	0.744	0.744	0.744
	COMM2	0.870	0.870	0.870	0.869	0.870	PA: 0.829	PA: 0.829	PA: 0.829	PA: 0.829	PA: 0.829					
	COMM3	0.867	0.866	0.867	0.867	0.867	CR: 0.897	CR: 0.897	CR: 0.897	CR: 0.897	CR: 0.897					
Conveni- ence	CONV1	0.855	0.856	0.857	0.890	0.889	CA: 0.834	CA: 0.834	CA: 0.834	CA: 0.766	CA: 0.766	0.750	0.750	0.750	0.810	0.810
	CONV2	0.873	0.870	0.866	–	–	PA: 0.835	PA: 0.835	PA: 0.835	PA: 0.771	PA: 0.772					
	CONV3	0.870	0.873	0.875	0.910	0.911	CR: 0.900	CR: 0.900	CR: 0.900	CR: 0.895	CR: 0.895					
Financial advan- tages	FA1	0.873	0.876	0.878	0.899	0.899	CA: 0.829	CA: 0.829	CA: 0.829	CA: 0.739	CA: 0.739	0.746	0.746	0.745	0.793	0.793
	FA2	0.879	0.882	0.887	–	–	PA: 0.832	PA: 0.835	PA: 0.841	PA: 0.742	PA: 0.742					
	FA3	0.838	0.832	0.823	0.882	0.882	CR: 0.898	CR: 0.898	CR: 0.898	CR: 0.885	CR: 0.885					
Product Variety	PV1	0.857	0.870	0.884	0.897	0.899	CA: 0.782	CA: 0.782	CA: 0.782	CA: 0.715	CA: 0.715	0.696	0.694	0.691	0.778	0.778
	PV2	0.813	0.797	0.777	–	–	PA: 0.789	PA: 0.803	PA: 0.832	PA: 0.722	PA: 0.725					
	PV3	0.832	0.831	0.829	0.867	0.864	CR: 0.873	CR: 0.872	CR: 0.870	CR: 0.875	CR: 0.875					

Table 1 (continued)

Con-struct	Item	Loading				Internal consistency reliability						Average variance extracted					
		RI (Mode A)	RI (Mode B)	Two stage	Hybrid (Mode A)	Hybrid (Mode B)	RI (Mode A)	RI (Mode B)	Two-stage	Hybrid (Mode A)	Hybrid (Mode B)	RI (Mode A)	RI (Mode B)	Two-stage	Hybrid (Mode A)	Hybrid (Mode B)	
Time Saving	TS1	0.882	0.881	0.880	0.916	0.914	CA: 0.857	CA: 0.857	CA: 0.857	CA: 0.791	CA: 0.791	0.777	0.777	0.777	0.827	0.827	
	TS2	0.887	0.888	0.891	–	–	PA: 0.858	PA: 0.858	PA: 0.859	PA: 0.793	PA: 0.792						
	TS3	0.876	0.876	0.873	0.903	0.905	CR: 0.913	CR: 0.913	CR: 0.913	CR: 0.905	CR: 0.905						
Enjoy-ment	Enjoy1	0.953	0.953	0.955	1.000	1.000	CA: 0.893	CA: 0.893	CA: 0.893	CA: 1.000	CA: 1.001	0.904	0.904	0.904	1.000	1.000	
	Enjoy2	0.949	0.948	0.946	–	–	PA: 0.894	PA: 0.895	PA: 0.898	PA: 1.000	PA: 1.001						
IPTO	Int1	0.906	0.905	0.905	0.906	0.903	CR: 0.949	CR: 0.949	CR: 0.949	CR: 1.000	CR: 1.001						
							CA: 0.757	CA: 0.757	CA: 0.757	CA: 0.757	CA: 0.757	0.805	0.805	0.805	0.805	0.805	
	Int2	0.888	0.889	0.889	0.887	0.891	PA: 0.761	PA: 0.760	PA: 0.760	PA: 0.761	PA: 0.759						
							CR: 0.892	CR: 0.892	CR: 0.892	CR: 0.892	CR: 0.892						

IPTO Intention to purchase travel online, *CA* Cronbach Alpha, P_A ; Rho_A , *CR* Composite reliability, *M-A* Mode A, *M-B* Mode B
The result of the two-stage from this Table 1 represents both Mode A and Mode B
The result of the repeated indicator approach (Mode A and Mode B) from this Table 1, will also represents the result of the Ringle's two-stage approach (stage one on both Mode A and Mode B) as well as the improved repeated indicator approach (Mode A and Mode B)

Table 2 HTMT-based discriminant validity using assessment

Approach	Construct	1	2	3	4	5	6	7
RI (Mode A)	1. Communicability							
	2. Convenience	0.528 [0.429;0.615]						
	3. Enjoyment	0.376 [0.275;0.474]	0.506 [0.397;0.607]					
	4. Financial advantages	0.377 [0.271;0.482]	0.628 [0.532;0.712]	0.622 [0.531;0.703]				
	5. Intention to purchase travel online	0.620 [0.516;0.709]	0.571 [0.486;0.654]	0.443 [0.342;0.539]	0.460 [0.356;0.553]			
	6. Product variety	0.431 [0.326;0.531]	0.603 [0.487;0.697]	0.573 [0.472;0.662]	0.636 [0.543;0.722]	0.430 [0.328;0.528]		
	7. Time saving	0.513 [0.413;0.607]	0.742 [0.664;0.811]	0.589 [0.493;0.678]	0.684 [0.614;0.747]	0.590 [0.490;0.672]	0.681 [0.589;0.763]	
RI (Mode B)	1. Communicability							
	2. Convenience	0.528 [0.434;0.616]						
	3. Enjoyment	0.376 [0.275;0.475]	0.506 [0.401;0.602]					
	4. Financial advantages	0.377 [0.266;0.483]	0.628 [0.539;0.710]	0.622 [0.527;0.702]				
	5. Intention to purchase travel online	0.620 [0.516;0.715]	0.571 [0.488;0.651]	0.443 [0.335;0.540]	0.460 [0.354;0.557]			
	6. Product variety	0.431 [0.330;0.536]	0.603 [0.490;0.700]	0.573 [0.471;0.662]	0.636 [0.541;0.719]	0.430 [0.330;0.529]		
	7. Time saving	0.513 [0.411;0.610]	0.742 [0.665;0.809]	0.589 [0.491;0.673]	0.684 [0.611;0.748]	0.590 [0.498;0.675]	0.681 [0.588;0.764]	

Table 2 (continued)

Approach	Construct	1	2	3	4	5	6	7
Two-stage (Mode A and Mode B)	1. Communica- bility							
	2. Convenience	0.528 [0.431;0.613]						
	3. Enjoyment	0.376 [0.272;0.472]	0.506 [0.404;0.606]					
	4. Financial advan- tages	0.377 [0.269;0.481]	0.628 [0.540;0.711]	0.622 [0.530;0.703]				
	5. Intention to purchase travel online	0.620 [0.518;0.714]	0.571 [0.483;0.650]	0.443 [0.339;0.538]	0.460 [0.349;0.554]			
	6. Product variety	0.431 [0.324;0.528]	0.603 [0.481;0.698]	0.573 [0.470;0.661]	0.636 [0.541;0.721]	0.430 [0.325;0.527]		
	7. Time saving	0.513 [0.413;0.606]	0.742 [0.662;0.808]	0.589 [0.491;0.676]	0.684 [0.610;0.749]	0.590 [0.496;0.673]	0.681 [0.590;0.673]	
Hybrid (Mode A)	1. Communica- bility							
	2. Convenience	0.533 [0.428;0.625]						
	3. Enjoyment	0.347 [0.245;0.435]	0.443 [0.344;0.544]					
	4. Financial advan- tages	0.358 [0.240;0.466]	0.584 [0.479;0.680]	0.611 [0.516;0.692]				
	5. Intention to purchase travel online	0.620 [0.518;0.710]	0.576 [0.480;0.668]	0.421 [0.325;0.509]	0.466 [0.344;0.575]			
	6. Product variety	0.460 [0.355;0.565]	0.645 [0.533;0.747]	0.565 [0.464;0.660]	0.683 [0.574;0.782]	0.505 [0.395;0.609]		

Table 2 (continued)

Approach	Construct	1	2	3	4	5	6	7
Hybrid (Mode B)	7. Time saving	0.510 [0.406;0.606]	0.764 [0.673;0.846]	0.531 [0.437;0.617]	0.708 [0.614;0.789]	0.579 [0.478;0.673]	0.723 [0.627;0.812]	
	1. Communicability							
	2. Convenience	0.533 [0.425;0.626]						
	3. Enjoyment	0.347 [0.245;0.437]	0.443 [0.337;0.540]					
	4. Financial advantages	0.358 [0.243;0.467]	0.584 [0.480;0.678]	0.611 [0.519;0.690]				
	5. Intention to purchase travel online	0.620 [0.516;0.712]	0.576 [0.479;0.664]	0.421 [0.327;0.507]	0.466 [0.348;0.577]			
	6. Product variety	0.460 [0.351;0.566]	0.645 [0.536;0.742]	0.565 [0.459;0.654]	0.683 [0.575;0.783]	0.505 [0.393;0.611]		
	7. Time saving	0.510 [0.403;0.609]	0.764 [0.674;0.844]	0.531 [0.433;0.614]	0.708 [0.616;0.794]	0.579 [0.476;0.678]	0.723 [0.625;0.812]	

The result of the two-stage from this Table 2 represents both Mode A and Mode B

The result of the repeated indicator approach (Mode A and Mode B) from this Table 1, will also represents the result of Ringle's two-stage approach (stage one on both Mode A and Mode B) as well as the improved repeated indicator approach (Mode A and Mode B)

Following this, it was necessary to investigate whether critical levels of collinearity affect the estimations of the HCM's formative construct/indicator weights. The results in Table 3 shows that all formative constructs/indicators yield VIF values below the threshold value of 3.3 (Diamantopoulos and Siguaw 2006). The construct/indicator weights' significance testing uses the 95% BCa bootstrap confidence intervals. The result shows that when Mode A was used on all four approaches, the weights of the LOCs are significant (see Table 3). In contrast, the repeated indicator (Mode B), Ringle et al.'s (2012) two-stage (Mode B), and the Becker et al.'s (2012) two-stage (Mode B) approaches provide insignificant results regarding weights of LOCs for both financial advantages and product variety. The other LOCs (i.e., convenience, time saving, and enjoyment) have significant weights. As for the hybrid approach (Mode B), all LOCs exhibit significant weights on PRA with the only exception of product variety. The nonsignificant weights of the LOCs' are not discarded in this study because they influence the content validity of the PRA construct (Hair et al. 2017a, Chapter 5). Therefore, all LOCs are maintained in the study because they are relevant to form HOCs of PRA.

The quality of HOC was also tested (see Hair et al. 2017b, Chapter 5). This step applied the quality criteria for formative items, but at the HOC level (Chin 2010), in which the LOCs now act as indicators of the HOC (Hair et al. 2017b). As for the case of the two-stage^a and two-stage^b (both Mode A and Mode B), each construct is represented by an indicator (the latent variable score) obtained in the first stage. The formative measurement model assessment initially focuses on the constructs' convergent validity by conducting a redundancy analysis of a particular formative construct (i.e., PRA). The redundancy analysis draws on the global single-item that summarizes the essence of each formative measured construct. Based on Table 3, all the path relationships between the formative measured construct and its global single-item measure (i.e., 0.750, 0.974, 0.769, 0.881, 0.765, 0.872, 0.816, and 0.972) are above the critical value of 0.70 (Hair et al. 2017b; Cheah et al. 2018).

Since the results of the reflective and formative measurement model assessment exhibit satisfactory levels of reliability and validity, the assessment of the structural model can be carried out. Following the structural model assessment procedure (Hair et al. 2017b, Chapter 6), the structural model for collinearity issues is assessed by examining the variance inflated factor (VIF) values of all predictor constructs in the model. As all VIF values are below the threshold of 3.3 (Diamantopoulos and Siguaw 2006), it can be concluded that collinearity is not a critical issue for all of the approaches (Table 4).

The path coefficient estimates of the structural model are estimated in Table 4. The improved repeated indicator approach (Mode A and Mode B) leads to higher path coefficients of the direct effect of COMM and PRA (Mode A: 0.517 and Mode B: 0.535) as well as PRA and IPTO (Mode A: 0.411 and Mode B: 0.448) than the other approaches. Specifically, the improved repeated indicator approach yields the highest coefficient of determination (R^2) values of IPTO (Mode A: 0.367 and Mode B: 0.385). It was also observed that the path coefficient results of the Ringle et al.'s (2012) two-stage and the Becker et al.'s (2012) two-stage approaches are identical for all direct relationships (i.e., COMM on PRA, COMM on IPTO, and PRA on IPTO). Interestingly, the use of Mode B in the repeated indicator approach produces a better path coefficient result of direct (COMM on PRA and PRA on COMM) and indirect effects compared to both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approaches (Mode B). In contrast, when Mode A was used in the repeated indicator approach, the result yields a consistent path coefficient with both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approaches (Mode A).

Table 3 Formative measurement model assessment results

HOC	LOCs	RI (Mode A)	RI (Mode B)	Outer weights and 95% BCa confidence interval					
				Two-stage (Mode A) ^a	Two-stage (Mode B) ^a	Two stage (Mode A) ^b	Two-stage (Mode B) ^b	Hybrid (Mode A)	Hybrid (Mode B)
PRA	CONV	0.275 [0.254; 0.297]	0.418 [0.242; 0.606]	0.293 [0.264; 0.327]	0.457 [0.291; 0.613]	0.386 [0.355; 0.425]	0.451 [0.287; 0.619]	0.175 [0.098; 0.251]	0.162 [0.048; 0.266]
	TS	0.302 [0.284; 0.323]	0.460 [0.238; 0.666]	0.297 [0.271; 0.327]	0.437 [0.252; 0.603]	0.242 [0.219; 0.265]	0.427 [0.248; 0.594]	0.269 [0.193; 0.343]	0.326 [0.227; 0.416]
	FA	0.261 [0.243; 0.279]	0.071 [−0.143; 0.259]	0.222 [0.192; 0.252]	0.019 [−0.162; 0.203]	0.233 [0.211; 0.256]	0.023 [−0.156; 0.195]	0.286 [0.223; 0.346]	0.235 [0.150; 0.327]
	ENJOY	0.194 [0.178; 0.211]	0.206 [0.016; 0.398]	0.226 [0.192; 0.257]	0.191 [0.035; 0.352]	0.197 [0.164; 0.227]	0.185 [0.023; 0.339]	0.257 [0.181; 0.330]	0.324 [0.194; 0.451]
	PV	0.229 [0.208; 0.249]	0.044 [−0.139; 0.241]	0.225 [0.194; 0.255]	0.101 [−0.068; 0.265]	0.242 [0.219; 0.265]	0.123 [−0.047; 0.285]	0.101 [0.034; 0.165]	0.033 [−0.054; 0.134]
				VIF					
HOC	LOCs	RI (Mode A)	RI (Mode B)	Two-stage (Mode A) ^a	Two-stage (Mode B) ^a	Two stage (Mode A) ^b	Two-stage (Mode B) ^b	Hybrid (Mode A)	Hybrid (Mode B)
PRA	CONV	1.901	1.902	1.822	1.822	1.827	1.827	1.742	1.741
	TS	2.204	2.207	2.151	2.151	2.161	2.161	2.046	2.044
	FA	1.830	1.826	1.832	1.832	1.819	1.819	1.715	1.715
	ENJOY	1.616	1.616	1.608	1.608	1.607	1.607	1.576	1.576
	PV	1.688	1.694	1.675	1.675	1.686	1.686	1.694	1.695

Table 3 (continued)

HOC	RI (Mode A)	RI (Mode B)	Redundancy analysis					
			Two-stage (Mode A) ^a	Two-stage (Mode B) ^a	Two stage (Mode A) ^b	Two-stage (Mode B) ^b	Hybrid (Mode A)	Hybrid (Mode B)
PRA	0.750	0.974	0.769	0.881	0.765	0.872	0.816	0.972

PRA Perceived relative advantages, *CONV* Convenience, *TS* Time saving, *FA* Financial advantages, *ENJOY* Enjoyment, *PV* Product variety, *RI* Repeated indicator note, *HOC* Higher order component, *LOCs* Lower order components

The result of the repeated indicator approach (Mode A and Mode B) from this Table 3, will also represents the improved repeated indicator approach (Mode A and Mode B)

^aThe two-stage (Mode A and B) is result from Ringle's et al. (2012) procedure

^bThe two-stage (Mode A and Mode B) is from Becker's et al. (2012) procedure

Table 4 Structural model assessment

Approach	Relationship	Path	95% BCa confidence interval			VIF	R ²	f ²	Q ²	VAF (%)
RI (Mode A)	COMM -> PRA	0.477	0.394	0.551	1.325	0.227		0.220		
	COMM -> IPTO	0.321	0.234	0.416	1.294		0.121			
	PRA -> IPTO	0.359	0.271	0.443	1.294	0.342	0.152	0.262		
	COMM -> PRA -> IPTO	0.171	0.125	0.217						34.756
RI (Mode B)	COMM -> PRA	0.494	0.415	0.566	1.325	0.244		0.235		
	COMM -> IPTO	0.300	0.221	0.402	1.322		0.106			
	PRA -> IPTO	0.389	0.285	0.456	1.322	0.357	0.178	0.267		
	COMM -> PRA -> IPTO	0.192	0.144	0.240						39.024
Two-stage (Mode A) ^a	COMM -> PRA	0.479	0.397	0.549	1.000	0.229		0.132		
	COMM -> IPTO	0.320	0.231	0.413	1.298		0.120			
	PRA -> IPTO	0.361	0.271	0.441	1.298	0.343	0.152	0.262		
	COMM -> PRA -> IPTO	0.173	0.129	0.220						35.091
Two-stage (Mode B) ^a	COMM -> PRA	0.493	0.404	0.561	1.000	0.243		0.131		
	COMM -> IPTO	0.308	0.219	0.405	1.322		0.110			
	PRA -> IPTO	0.375	0.279	0.452	1.322	0.349	0.163	0.265		
	COMM -> PRA -> IPTO	0.185	0.137	0.232						37.525
Two-stage (Mode A) ^b	COMM -> PRA	0.481	0.396	0.550	1.000	0.231		0.133		
	COMM -> IPTO	0.318	0.222	0.406	1.301		0.118			
	PRA -> IPTO	0.364	0.274	0.448	1.301	0.345	0.155	0.263		
	COMM -> PRA -> IPTO	0.175	0.130	0.224						35.497
Two-stage (Mode B) ^b	COMM -> PRA	0.495	0.411	0.566	1.000	0.245		0.133		
	COMM -> IPTO	0.306	0.219	0.402	1.324		0.109			
	PRA -> IPTO	0.376	0.283	0.456	1.324	0.350	0.164	0.265		
	COMM -> PRA -> IPTO	0.186	0.138	0.233						37.805
Hybrid (Mode A)	COMM -> PRA	0.452	0.364	0.525	1.314	0.204		0.196		
	COMM -> IPTO	0.353	0.263	0.444	1.257		0.146			
	PRA -> IPTO	0.308	0.220	0.393	1.257	0.318	0.111	0.244		
	COMM -> PRA -> IPTO	0.139	0.096	0.183						28.252
Hybrid (Mode B)	COMM -> PRA	0.454	0.370	0.529	1.314	0.206		0.199		
	COMM -> IPTO	0.342	0.249	0.433	1.260		0.138			
	PRA -> IPTO	0.332	0.236	0.415	1.260	0.330	0.130	0.252		
	COMM -> PRA -> IPTO	0.151	0.105	0.197						30.629
IRI (Mode A)	COMM -> PRA	0.517	0.427	0.595	1.325	0.267		0.220		

Table 4 (continued)

Approach	Relationship	Path	95% BCa confidence interval			VIF	R ²	f ²	Q ²	VAF (%)
IRI (Mode B)	COMM -> IPTO	0.280	0.187	0.383	1.365			0.091		
	PRA -> IPTO	0.411	0.301	0.501	1.365	0.367	0.195	0.262		
	COMM -> PRA	0.213	0.156	0.277						43.205
	-> IPTO									
	COMM -> PRA	0.535	0.446	0.614	1.325	0.287			0.235	
	COMM -> IPTO	0.253	0.162	0.363	1.402		0.074			
	PRA -> IPTO	0.448	0.330	0.526	1.402	0.385	0.233	0.267		
	COMM -> PRA	0.240	0.175	0.305						48.682
	-> IPTO									

COMM Communicability, *PRA* Perceived relative advantages, *IPTO* Intention to purchase travel online, *RI* Repeated indicator, *IRI* Improved repeated indicator approach

^aThe two-stage (Mode A and B) is result from Ringle's et al. (2012) procedure

^bThe two-stage (Mode A and Mode B) is from Becker's et al. (2012) procedure

Notably, when the repeated indicator, hybrid, and the improved repeated indicator approaches (both Mode A and Mode B) were used on the relationship between COMM and PRA, the variance of the HOC was perfectly explained by its LOCs. The explained variance of PRA is equal to 1 for the repeated indicators approach (Mode A and Mode B), whereas it shows inflated value for the hybrid approach (Mode A and Mode B), and a value of more than 1 for the improved repeated indicator approach (Mode A and Mode B) (Ringle et al. 2012). In other words, these approaches will produce an artificial and perfect R² for the latent endogenous variable and can also cause artificially correlated residuals. One solution to precisely estimate or yield a reasonable R² value from these approaches is to use the generated latent variable score of the HOC (i.e., only use the PRA's latent variable score without or deleting the LOCs from the model). Based on this procedure, the use of improved repeated indicator with Mode A and Mode B obtained a better result of R² (Mode A: 0.267 and Mode B: 0.287) than the conventional repeated indicator with Mode A (0.227) and Mode B (0.244). Moreover, it seems that repeated indicator approach with Mode B (R²=0.244) yield a similar result when compared to Ringle et al.'s (2012) two-stage approach with Mode B (R²=0.243) as well as Becker et al.'s Mode B (R²=0.245) (similar condition occurs with Mode A estimation). Lastly, it seems that the hybrid approach (Mode A and Mode B) also produces similar R² value (0.204 and 0.206) for the relationship between COMM and PRA (see Table 4).

This study further examines the mediation effect (Baron and Kenny 1986; Ingusci et al. 2016). Specifically, PRA serves as the mediator in the relationship between the COMM and IPTO. Based on Table 4, all the approaches yield a significant relationship for the indirect effects ($p < 0.01$), the improved repeated indicator with Mode B yields the highest path coefficient value (0.240), followed by the improved repeated indicator with Mode A (0.213), the repeated indicator with Mode B (0.192), Becker et al.'s (2012) two-stage with Mode B (0.186), Ringle et al.'s (2012) two-stage with Mode B (0.185), Becker et al.'s (2012) two-stage with Mode A (0.175), Ringle et al.'s (2012) two-stage with Mode A (0.173), the repeated indicator with Mode A (0.171), hybrid with Mode B (0.151), and lastly the hybrid with Mode A (0.139). In all the approaches significant

($p < 0.01$) direct effects between the COMM and IPTO are also found. These results suggest that the PRA partially mediates (complementary) the relationship between the COMM and IPTO (Nitzl et al. 2016; Zhao et al. 2010). In addition, the variance accounted for (VAF) was analyzed. Technically, the VAF represents the ratio of the indirect effect to the total effect in the mediator model. The use of VAF is able to help determine how much of the target construct's variance is explained by the indirect relationship via the mediator variable. The VAF value in Table 4 shows that the improved repeated indicator with Mode B yields the highest VAF value (48.682%), followed by the improved repeated indicator with Mode A (43.205%), the repeated indicator with Mode B (39.024%), the Becker et al.'s (2012) two-stage with Mode B (37.805%), the Ringle et al.'s (2012) two-stage with Mode B (37.525%), the Becker et al.'s (2012) two-stage with Mode A (35.497%), the Ringle et al.'s (2012) two-stage with Mode A (35.091%), the repeated indicator with Mode A (34.756%), the hybrid with Mode B (30.629%), and lastly the hybrid with Mode A (28.252%).

Table 4 also shows the f^2 effect sizes result. According to Cohen (1988), weak effect sizes (f^2) occur for all the approaches (0.074–0.146) in the relationship between Communicability and PRA, which the lowest f^2 on this relationship is used as the improved repeated indicator with Mode B (0.074). However, the relationship between COMM and PRA exhibits medium effect size (f^2) when for all the approaches (0.111–0.233), which the highest f^2 on this relationship is used as the improved repeated indicator with Mode B (0.233).

There are also substantial differences regarding the result of cross-validated redundancy when Q^2 is assessed for the dependent variable (Geisser 1974; Stone 1974) (Table 4). The predictive relevance obtained when using the conventional repeated indicator approach (Mode A and Mode B) and the improved repeated indicator approach (Mode A and Mode B) is identical for the relationship between COMM and PRA. Additionally, the predictive relevance result of COMM and PRA also exhibits similar results for both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approaches. However, the predictive relevance of the relationship between PRA and IPTO yields almost identical results for all the approaches regardless of whether Mode A or Mode B is used (0.262–0.267), except for the hybrid approaches (Mode A and Mode B). Overall, all approaches exhibit a predictive relevance result because the Q^2 value is greater than 0.

This study further examines the predictive performance of each approach by using PLSpredict (see Shmueli et al. 2016). The method splits the data into a training and hold-out sample to evaluate the predictive performance of both indicator and construct predictions in the PLSPM. Based on the Table 5 result, it can be observed that all approaches show that the PLS-PM results provide a lower prediction error (e.g., in terms of RMSE) and greater Q^2 values, than the LM outcomes. This indicates that a theoretically established path model improves (or at least does not worsen) the predictive performance of the available indicator data.

In addition, the PLSpredict results on latent variable exhibits that the Q^2 value of the conventional repeated indicator (Mode B), Ringle et al.'s (2012) two-stage, and Becker et al.'s (2012) two-stage, as well as the improved repeated indicator (Mode B) have a stronger prediction performance than other approaches (Table 6). This finding is alike with Becker et al. (2013)' study who substantiated the higher predictive capabilities of Mode B model estimations. In contrast, all other HCM approaches in this study did not result in satisfactory predictions of construct. As indicated in Table 6, the PRA latent prediction exhibits a Q^2 value of negative, in which the prediction error of the PLS-PM results is larger than the error of simply using the mean values.

Table 5 Manifest variable prediction

Approach		PLS			LM			PLS-LM					
		RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²
RI (Mode A)	CONV1	0.990	0.764	15.979	0.134	0.995	0.768	16.074	0.125	-0.005	-0.004	-0.095	0.009
	CONV2	0.997	0.767	16.825	0.135	1.003	0.772	16.975	0.125	-0.006	-0.005	-0.150	0.010
	CONV3	1.035	0.807	17.738	0.148	1.041	0.812	17.833	0.137	-0.006	-0.005	-0.095	0.011
	Enjoy1	1.158	0.949	22.852	0.096	1.163	0.947	22.822	0.088	-0.005	0.002	0.030	0.008
	Enjoy2	1.154	0.935	22.680	0.086	1.157	0.935	22.686	0.081	-0.003	0.000	-0.006	0.005
	FA1	1.124	0.899	20.959	0.071	1.130	0.904	21.059	0.061	-0.006	-0.005	-0.100	0.010
	FA2	1.096	0.879	20.072	0.090	1.104	0.881	20.134	0.077	-0.008	-0.002	-0.062	0.013
	FA3	1.081	0.871	18.630	0.045	1.087	0.882	18.863	0.035	-0.006	-0.011	-0.233	0.010
	PV1	0.986	0.789	16.049	0.120	0.991	0.783	15.894	0.111	-0.005	0.006	0.155	0.009
	PV2	1.147	0.915	21.565	0.053	1.152	0.923	21.739	0.046	-0.005	-0.008	-0.174	0.007
	PV3	1.102	0.896	19.769	0.067	1.106	0.904	19.991	0.060	-0.004	-0.008	-0.222	0.007
	TS1	1.037	0.791	17.758	0.143	1.043	0.794	17.843	0.132	-0.006	-0.003	-0.085	0.011
	TS2	0.929	0.722	14.352	0.148	0.935	0.728	14.455	0.136	-0.006	-0.006	-0.103	0.012
RI (Mode B)	TS3	1.018	0.817	16.205	0.126	1.023	0.821	16.285	0.118	-0.005	-0.004	-0.080	0.008
	Int1	1.005	0.790	16.754	0.213	1.009	0.790	16.712	0.207	-0.004	0.000	0.042	0.006
	Int2	0.971	0.762	15.530	0.168	0.975	0.767	15.637	0.161	-0.004	-0.005	-0.107	0.007
	CONV1	0.989	0.763	15.960	0.135	0.995	0.767	16.056	0.124	-0.006	-0.004	-0.096	0.011
	CONV2	0.997	0.767	16.825	0.136	1.003	0.773	16.988	0.124	-0.006	-0.006	-0.163	0.012
	CONV3	1.035	0.807	17.739	0.148	1.041	0.812	17.828	0.137	-0.006	-0.005	-0.089	0.011
	Enjoy1	1.158	0.949	22.841	0.096	1.160	0.946	22.783	0.092	-0.002	0.003	0.058	0.004
	Enjoy2	1.154	0.935	22.666	0.086	1.155	0.935	22.659	0.084	-0.001	0.000	0.007	0.002
	FA1	1.123	0.899	20.952	0.071	1.129	0.903	21.054	0.063	-0.006	-0.004	-0.102	0.008
	FA2	1.097	0.880	20.082	0.090	1.105	0.881	20.147	0.077	-0.008	-0.001	-0.065	0.013
	FA3	1.081	0.870	18.626	0.045	1.087	0.882	18.874	0.034	-0.006	-0.012	-0.248	0.011
	PV1	0.985	0.786	15.990	0.121	0.989	0.781	15.855	0.114	-0.004	0.005	0.135	0.007

Table 5 (continued)

Approach	PLS				LM				PLS-LM			
	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²
PV2	1.147	0.916	21.570	0.054	1.151	0.924	21.735	0.047	-0.004	-0.008	-0.165	0.007
PV3	1.101	0.896	19.758	0.068	1.105	0.903	19.980	0.063	-0.004	-0.007	-0.222	0.005
TS1	1.036	0.790	17.737	0.143	1.041	0.793	17.804	0.135	-0.005	-0.003	-0.067	0.008
TS2	0.928	0.721	14.336	0.150	0.933	0.725	14.403	0.141	-0.005	-0.004	-0.067	0.009
TS3	1.016	0.816	16.176	0.128	1.020	0.819	16.251	0.121	-0.004	-0.003	-0.075	0.007
Int1	1.005	0.789	16.734	0.213	1.009	0.788	16.682	0.207	-0.004	0.001	0.052	0.006
Int2	0.971	0.762	15.516	0.167	0.976	0.767	15.633	0.159	-0.005	-0.005	-0.117	0.008
Int1	1.005	0.789	16.738	0.214	1.007	0.788	16.688	0.210	-0.002	0.001	0.050	0.004
Int2	0.971	0.762	15.528	0.168	0.975	0.767	15.632	0.161	-0.004	-0.005	-0.104	0.007
CONV	0.905	0.702	126.503	0.186	0.908	0.704	134.045	0.179	-0.003	-0.002	-7.542	0.007
ENJOY	0.949	0.761	108.203	0.102	0.951	0.761	105.076	0.099	-0.002	0.000	3.127	0.003
FA	0.955	0.756	408.012	0.091	0.959	0.761	350.224	0.083	-0.004	-0.005	57.788	0.008
PV	0.940	0.775	156.366	0.119	0.944	0.780	155.138	0.112	-0.004	-0.005	1.228	0.007
TS	0.906	0.712	245.119	0.182	0.910	0.715	262.499	0.175	-0.004	-0.003	-17.380	0.007
Int1	1.004	0.789	16.736	0.213	1.007	0.789	16.694	0.209	-0.003	0.000	0.042	0.004
Int2	0.971	0.762	15.522	0.168	0.975	0.767	15.628	0.161	-0.004	-0.005	-0.106	0.007
CONV	0.904	0.700	132.716	0.187	0.907	0.703	133.751	0.180	-0.003	-0.003	-1.035	0.007
ENJOY	0.949	0.763	107.101	0.102	0.952	0.761	105.151	0.098	-0.003	0.002	1.950	0.004
FA	0.954	0.757	364.846	0.094	0.958	0.760	342.328	0.086	-0.004	-0.003	22.518	0.008
PV	0.941	0.778	150.853	0.119	0.944	0.780	155.211	0.114	-0.003	-0.002	-4.358	0.005
TS	0.906	0.710	257.962	0.183	0.911	0.716	261.245	0.174	-0.005	-0.006	-3.283	0.009
Int1	1.004	0.789	16.723	0.214	1.008	0.788	16.674	0.208	-0.004	0.001	0.049	0.006
Int2	0.971	0.762	15.519	0.167	0.976	0.767	15.636	0.159	-0.005	-0.005	-0.117	0.008
CONV	0.904	0.702	119.490	0.186	0.909	0.705	126.825	0.177	-0.005	-0.003	-7.335	0.009

Table 5 (continued)

Approach	PLS				LM				PLS-LM				
	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	
Two-stage (Mode B) ^b	ENJOY	0.949	0.761	108.938	0.102	0.952	0.762	105.507	0.097	-0.003	-0.001	3.431	0.005
	FA	0.955	0.754	567.073	0.092	0.958	0.759	494.270	0.085	-0.003	-0.005	72.803	0.007
	PV	0.938	0.775	135.516	0.123	0.942	0.780	133.653	0.116	-0.004	-0.005	1.863	0.007
	TS	0.906	0.712	179.127	0.182	0.911	0.715	189.679	0.173	-0.005	-0.003	-10.552	0.009
	Int1	1.004	0.789	16.718	0.214	1.008	0.788	16.679	0.208	-0.004	0.001	0.039	0.006
	Int2	0.971	0.762	15.520	0.168	0.976	0.767	15.635	0.159	-0.005	-0.005	-0.115	0.009
	CONV	0.904	0.701	124.962	0.187	0.908	0.705	126.656	0.179	-0.004	-0.004	-1.694	0.008
	ENJOY	0.949	0.763	107.560	0.102	0.951	0.761	105.463	0.099	-0.002	0.002	2.097	0.003
	FA	0.953	0.755	509.154	0.096	0.958	0.759	497.386	0.086	-0.005	-0.004	11.768	0.010
	PV	0.939	0.777	132.161	0.122	0.942	0.779	133.582	0.116	-0.003	-0.002	-1.421	0.006
Hybrid (Mode A)	TS	0.906	0.710	186.339	0.182	0.911	0.715	189.294	0.174	-0.005	-0.005	-2.955	0.008
	CONV1	0.989	0.763	15.961	0.136	0.993	0.767	16.037	0.128	-0.004	-0.004	-0.076	0.008
	CONV2	0.996	0.774	16.968	0.137	1.004	0.773	16.983	0.124	-0.008	0.001	-0.015	0.013
	CONV3	1.034	0.807	17.713	0.149	1.040	0.811	17.816	0.139	-0.006	-0.004	-0.103	0.010
	Enjoy1	1.158	0.948	22.833	0.096	1.163	0.947	22.838	0.089	-0.005	0.001	-0.005	0.007
	Enjoy2	1.153	0.932	22.604	0.088	1.156	0.935	22.681	0.083	-0.003	-0.003	-0.077	0.005
	FA1	1.124	0.901	21.013	0.071	1.130	0.904	21.074	0.060	-0.006	-0.003	-0.061	0.011
	FA2	1.095	0.871	19.865	0.093	1.102	0.880	20.105	0.080	-0.007	-0.009	-0.240	0.013
	FA3	1.080	0.872	18.663	0.046	1.085	0.881	18.840	0.038	-0.005	-0.009	-0.177	0.008
	PV1	0.985	0.784	15.944	0.123	0.988	0.781	15.864	0.116	-0.003	0.003	0.080	0.007
PV2	1.146	0.916	21.572	0.056	1.151	0.924	21.745	0.048	-0.005	-0.008	-0.173	0.008	
PV3	1.104	0.894	19.734	0.065	1.106	0.905	20.010	0.060	-0.002	-0.011	-0.276	0.005	
TS1	1.036	0.790	17.751	0.145	1.041	0.793	17.805	0.137	-0.005	-0.003	-0.054	0.008	
TS2	0.929	0.722	14.367	0.149	0.934	0.726	14.429	0.140	-0.005	-0.004	-0.062	0.009	

Table 5 (continued)

Approach	PLS				LM				PLS-LM			
	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²
Hybrid (Mode B)	TS3	1.017	0.817	16.205	0.128	1.022	0.821	16.292	0.119	0.005	0.004	0.009
	Int1	1.004	0.789	16.731	0.214	1.006	0.787	16.658	0.212	0.002	0.002	0.002
	Int2	0.971	0.762	15.528	0.167	0.975	0.767	15.628	0.161	0.004	0.005	0.006
	CONV1	0.988	0.762	15.947	0.135	0.993	0.766	16.022	0.127	0.005	0.004	0.008
	CONV2	0.997	0.777	17.042	0.135	1.002	0.772	16.978	0.126	0.005	0.005	0.009
	CONV3	1.034	0.807	17.716	0.148	1.041	0.812	17.826	0.137	0.007	0.005	0.011
	Enjoy1	1.158	0.948	22.816	0.096	1.161	0.946	22.803	0.091	0.003	0.002	0.005
	Enjoy2	1.153	0.930	22.554	0.086	1.155	0.934	22.655	0.084	0.002	0.004	0.002
	FA1	1.123	0.901	21.008	0.071	1.129	0.903	21.048	0.062	0.006	0.002	0.009
	FA2	1.094	0.874	19.927	0.093	1.103	0.880	20.105	0.079	0.009	0.006	0.014
	FA3	1.081	0.873	18.672	0.046	1.086	0.882	18.857	0.037	0.005	0.009	0.009
	PV1	0.984	0.783	15.922	0.122	0.989	0.781	15.871	0.113	0.005	0.002	0.009
IRI (Mode A)	PV2	1.147	0.923	21.701	0.056	1.151	0.924	21.738	0.048	0.004	0.001	0.008
	PV3	1.103	0.894	19.716	0.066	1.105	0.904	19.990	0.062	0.002	0.010	0.004
	TS1	1.035	0.789	17.737	0.145	1.040	0.792	17.799	0.136	0.005	0.003	0.009
	TS2	0.928	0.721	14.339	0.150	0.934	0.727	14.433	0.139	0.006	0.006	0.011
	TS3	1.016	0.816	16.186	0.128	1.020	0.820	16.261	0.121	0.004	0.004	0.007
	Int1	1.004	0.789	16.726	0.215	1.006	0.788	16.663	0.212	0.002	0.001	0.003
	Int2	0.970	0.761	15.508	0.169	0.974	0.766	15.616	0.162	0.004	0.005	0.007
	CONV1	0.989	0.763	15.964	0.135	0.994	0.766	16.039	0.127	0.005	0.003	0.008
	CONV2	0.996	0.766	16.811	0.136	1.003	0.772	16.982	0.125	0.007	0.006	0.011
	CONV3	1.035	0.807	17.736	0.148	1.042	0.812	17.833	0.136	0.007	0.005	0.012
	Enjoy1	1.158	0.949	22.853	0.096	1.163	0.947	22.834	0.089	0.005	0.002	0.007
	Enjoy2	1.153	0.935	22.670	0.087	1.155	0.934	22.667	0.084	0.002	0.001	0.003

Table 5 (continued)

Approach	PLS				LM				PLS-LM				
	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	
IRI (Mode B)	FA1	1.124	0.899	20.972	0.071	1.130	0.904	21.067	0.062	-0.006	-0.005	-0.095	0.009
	FA2	1.097	0.879	20.066	0.090	1.104	0.880	20.127	0.078	-0.007	-0.001	-0.061	0.012
	FA3	1.081	0.870	18.624	0.045	1.088	0.883	18.881	0.033	-0.007	-0.013	-0.257	0.012
	PV1	0.985	0.789	16.043	0.121	0.989	0.781	15.863	0.114	-0.004	0.008	0.180	0.007
	PV2	1.146	0.915	21.555	0.055	1.150	0.923	21.714	0.048	-0.004	-0.008	-0.159	0.007
	PV3	1.102	0.896	19.769	0.067	1.105	0.904	19.994	0.061	-0.003	-0.008	-0.225	0.006
	TS1	1.035	0.789	17.726	0.145	1.041	0.793	17.804	0.135	-0.006	-0.004	-0.078	0.010
	TS2	0.928	0.721	14.334	0.150	0.933	0.726	14.421	0.141	-0.005	-0.005	-0.087	0.009
	TS3	1.016	0.815	16.171	0.128	1.020	0.819	16.246	0.120	-0.004	-0.004	-0.075	0.008
	Int1	1.005	0.789	16.736	0.214	1.009	0.789	16.711	0.207	-0.004	0.000	0.025	0.007
	Int2	0.971	0.762	15.525	0.168	0.975	0.767	15.639	0.161	-0.004	-0.005	-0.114	0.007
	CONV1	0.989	0.763	15.968	0.136	0.994	0.767	16.046	0.127	-0.005	-0.004	-0.078	0.009
	CONV2	0.996	0.766	16.815	0.137	1.002	0.772	16.965	0.127	-0.006	-0.006	-0.150	0.010
	CONV3	1.034	0.807	17.729	0.150	1.040	0.811	17.824	0.140	-0.006	-0.004	-0.095	0.010
	Enjoy1	1.158	0.949	22.851	0.097	1.162	0.947	22.818	0.090	-0.004	0.002	0.033	0.007
	Enjoy2	1.154	0.935	22.681	0.088	1.156	0.935	22.680	0.085	-0.002	0.000	0.001	0.003
	FA1	1.124	0.899	20.961	0.072	1.130	0.904	21.076	0.062	-0.006	-0.005	-0.115	0.010
	FA2	1.097	0.880	20.086	0.091	1.105	0.881	20.152	0.078	-0.008	-0.001	-0.066	0.013
	FA3	1.081	0.870	18.621	0.046	1.086	0.882	18.855	0.037	-0.005	-0.012	-0.234	0.009
	PV1	0.985	0.787	15.998	0.122	0.989	0.782	15.884	0.114	-0.004	0.005	0.114	0.008
	PV2	1.147	0.916	21.570	0.055	1.151	0.924	21.743	0.047	-0.004	-0.008	-0.173	0.008
	PV3	1.101	0.895	19.758	0.068	1.105	0.904	19.998	0.061	-0.004	-0.009	-0.240	0.007
	TS1	1.035	0.789	17.734	0.145	1.041	0.792	17.805	0.135	-0.006	-0.003	-0.071	0.010
	TS2	0.929	0.721	14.338	0.151	0.934	0.726	14.425	0.141	-0.005	-0.005	-0.087	0.010

Table 5 (continued)

Approach	PLS				LM				PLS-LM			
	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²	RMSE	MAE	MAPE	Q ²
TS3	1.016	0.816	16.176	0.130	1.021	0.819	16.257	0.122	−0.005	−0.003	−0.081	0.008
Int1	1.005	0.790	16.747	0.214	1.009	0.790	16.707	0.208	−0.004	0.000	0.040	0.006
Int2	0.971	0.762	15.522	0.168	0.975	0.766	15.624	0.161	−0.004	−0.004	−0.102	0.007

PRA Perceived relative advantages, *CONV* Convenience, *TS* Time saving, *FA* Financial advantages, *ENJOY* Enjoyment, *PV* Product variety

^aThe two-stage (Mode A and B) is result from Ringle's et al. (2012) procedure

^bThe two-stage (Mode A and Mode B) is from Becker's et al. (2012) procedure

Table 6 Latent variable prediction

Approach	Construct	RMSE	MAE	Q ²
RI (Mode A)	IPTO	0.501	0.366	0.009
	PRA	0.494	0.377	−0.028
RI (Mode B)	IPTO	0.498	0.365	0.017
	PRA	0.504	0.383	0.007
Two-stage (Mode A) ^a	IPTO	0.501	0.366	0.017
	PRA	0.492	0.376	−0.014
Two-stage (Mode B) ^a	IPTO	0.501	0.364	0.013
	PRA	0.509	0.384	0.001
Two-stage (Mode A) ^b	IPTO	0.503	0.367	0.015
	PRA	0.495	0.377	−0.013
Two-stage (Mode B) ^b	IPTO	0.501	0.366	0.009
	PRA	0.508	0.385	0.004
Hybrid (Mode A)	IPTO	0.500	0.363	0.012
	PRA	0.477	0.361	−0.063
Hybrid (Mode B)	IPTO	0.500	0.364	0.014
	PRA	0.483	0.365	−0.068
IRI (Mode A)	IPTO	0.501	0.366	0.024
	PRA	0.490	0.375	−0.014
IRI (Mode B)	IPTO	0.503	0.364	0.014
	PRA	0.505	0.382	0.004

PRA Perceived relative advantages, *IPTO* Intention to purchase travel online

^aThe two-stage (Mode A and B) is result from Ringle's et al. (2012) procedure

^bThe two-stage (Mode A and Mode B) is from Becker's et al. (2012) procedure

5 Discussions

The empirical example provided illustrates that all approaches have significant differences. The key difference is related to the weights of the LOCs on the HOC, as well as to the redundancy results. Another visible difference is regarding the path-coefficient results (direct and mediator result), inner VIF, R^2 , f^2 , and predictive relevance of the model, which are considerably higher when using the improved repeated indicator approach in both Mode A and Mode B. It implies that the choice of method can have a profound impact on the outcomes and conclusions that might be drawn from the estimates.

The repeated indicators approach using the total effect analysis in HCMs is easy to apply in both Mode A and Mode B. Specifically, the result of path coefficient produces from Mode A and Mode B repeated indicator approach is identical with Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approach (Mode A and Mode B) in the structural model assessment. In addition, the empirical example in this study also shows that the LOCs' weights result produced from the repeated indicator approach (Mode A and Mode B) are identical to both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approach (Mode A and Mode B). This shows that this approach is able to provide a precise

estimation of weights and the path coefficient result. In conclusion, it demonstrates both construct and predictive validity (Becker et al. 2012; Ringle et al. 2012).

Interestingly, the presence of an unequal numbers of indicators does not substantially influence the result of repeated indicator approach with Mode A and Mode B. For example, the path coefficient result from the predictor (COMM) on HOC (PRA) is relatively comparable to Ringle et al.'s (2012) and Becker et al.'s (2012) two stage approach (Mode A and B) result. In addition, when LOCs have unequal numbers of indicators the hybrid approaches (Mode A and Mode B) tend to overestimate the weight of the LOCs which have a larger number of indicators than the other constructs (see Table 4 for the weight result of FA and PV on PRA). This corroborates findings made by Wilson and Henseler (2007) as well as Becker et al. (2012) that the weights of the hybrid approaches (Mode A and Mode B) are biased regardless of equal or unequal numbers of indicators when assessing the reflective–formative type of HCM.

When using the hybrid approach (Mode A and Mode B) the assessment of the LOCs' reliability and validity is only possible when there are enough indicators to split them in half, and when there are at least two indicators for each LOC. Otherwise, it is not possible to estimate the LOCs' internal consistency reliability and validity (Dolbier et al. 2005; Wanous and Reichers 1996). An alternative is to apply the formula for correction of attenuation. However, it is considered as an inadequate substitute (Wanous and Reichers 1996). This highlights one of the hybrid approaches' disadvantages pertaining to the fact that more indicators are needed to properly evaluate the LOCs. As a consequence this often means that a survey needs to be prepare lengthier. Furthermore, when the number of indicators available to measure the LOCs is uneven, there are difficulties when deciding how to split the indicators between the first order and the second order constructs. In addition, the R^2 obtained from the hybrid approach (Mode A and Mode B) seems to differ from the others especially from the predictor (COMM) on HCM (PRA). Therefore, the use of the hybrid approach is not suggested (Mode A and Mode B) in actual practice.

With regards to Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approaches (Mode A and Mode B), they have the advantage of not requiring an equal number of indicators for the LOCs and yet provide reliable results. The approach produces an accurate R^2 result when compared to others approach where R^2 is equal to 1 (Ringle et al. 2012). Therefore, this approach is recommended when the HOC is an endogenous or a mediator. Furthermore, the use of different modes in the two-stage approach plays an important role in the explanatory result of path-coefficient. The two-stage approach with Mode A produces a stronger direct effect than Mode B. Conversely, the two-stage approach with Mode B produces a stronger indirect relationship than Mode A. Importantly, the estimation of two-stage approach using Mode B produces a similar or slightly improved in-sample predictive power (R^2) than using Mode A. This result was consistent with Becker et al.'s (2013) as well as Sarstedt et al.'s (2016) work that Mode A and Mode B estimations (no strong collinearity exist between the indicators) do not differ in terms of parameter accuracy when sample size is large. The out-of-sample prediction (see the latent prediction result of PRA and IPTO) of Mode B estimation in two-stage provides a similar result to Mode A. The result is also similar with Becker et al.'s (2013) as well as Sarstedt et al.'s (2016) work that when large sample sizes and large R^2 values occur in a study, both Mode A and Mode B perform equally well in terms of out-of-sample prediction.

Finally, the improved repeated indicator approach with Mode A and Mode B (an extension of conventional repeated indicator approach with Mode A and Mode B) presents stronger path coefficients (i.e., COMM on PRA and PRA on IPTO) result and a higher R^2 , when compared with the other approaches. Similar to the conventional repeated indicator approach with Mode A and Mode B as well as two-stage approach of Mode A and Mode B, it has the advantage of not requiring an equal number of indicators for the LOCs, and can be used when HOC is endogenous. Regarding the explained variance of the outcome construct (IPTO), the improved repeated indicator approach with Mode A and B produce the highest in-sample predictive power (R^2) and out of sample prediction (PLSpredict). The reason is the average reliability (based on the result produced by LOCs) was pre-specified in the composite estimation of the HOC. As a result, it enhances the composite estimations of the HOC's model parameters.

From a practical point of view, this study suggests that despite the method used to evaluate HOCs, researchers should expect to get different results. Therefore, the choice of the method must be made in consideration of factors related to the underlying theory and the research objectives. The choice is also dependent on the nature of the HOC and its relationship with the LOCs. First, it is suggested that researchers use the repeated indicators approach with Mode B, as it produces an identical result to both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approach with Mode B, without requiring an equal number of indicators from the LOCs. Subsequently, in order to produce a more efficient result of HCM, it is encouraged that researchers extend the use of conventional repeated indicator approach (Mode A or Mode B) to improved repeated indicator approach (especially Mode B). For the hybrid approach (Mode A and Mode B), its major pitfall is that it reduces the reliability of the measures by using only half the number of indicators for each construct. Such a scenario is common in most applied research if past instruments of a HCM are adopted or adapted. Therefore, it is not recommended that researchers use this approach in their studies. Finally, both Ringle et al.'s (2012) and Becker et al.'s (2012) two-stage approach (Mode A and Mode B) has the advantage of estimating a more parsimonious model at a higher level, it is also easy to apply the software of PLS. However, splitting the estimation of the two-stage approach into two phases that do not take into account the whole nomological network may confound the interpretation of results (Wilson and Henseler 2007). As a result, it "loses" information in the estimation of the network and causes weaker predictive relevance. Therefore, researchers should consider the trade-off of using the two-stage approach.

6 Conclusions

In summary, this study contributes to the PLS-PM literature by providing empirical evidence through the comparison of different approaches for estimating HCMs and using both modes (Modes A and B). Guidelines are provided on how to assess the reflective–formative second-order constructs that is either an endogenous construct or as a mediator. It is hoped that these efforts, as called for by Ali et al. (2018) and do Valle and Assaker (2016),

will encourage researchers from tourism or hospitality management field to operate the appropriate HCM techniques in their work. Based on the findings, this study encourages the use of improved repeated indicator approach (Mode B) in either tourism or hospitality management field. This study does have some limitations. For example, it would be more appropriate to compare the hybrid method with the other approaches if there were four indicators for each first order construct.

Moving forward, researchers should consider alternative estimation methods of PLS to gain a better insight into studies related to performance evaluation of HCM in the tourism setting. For instance, future research should be focused toward the usefulness of the other types of HCM (i.e., formative–formative type model). Also, there is a lack of research focusing on the performance of HCM in PLS-PM when moderation models are involved (Wilson 2010). Moderation analysis in PLS-PM is usually carried out with three different approaches that may yield different results depending on how the hierarchical latent variables are estimated: (1) a product-indicator approach, (2) a two-stage approach, (3) orthogonalized approach (see Hair et al. 2017b, Chapter 7). It is unclear how a combination of these approaches for moderation and HCM estimation would perform. Therefore, future research should consider investigating the performance of possible combination of HCM and moderator estimations, which may produce interesting results.

Furthermore, future studies may consider the use of qualitative external information when examining the HCM (Ciavolino et al. 2015). Such an approach would complement the HCM model parameter's estimation as it characterizes the observations according to the context of study as well as identifies potential heterogeneity of the target population. Finally, some new estimators in PLS such as the nonlinear principal component analysis (NPCA), Rasch analysis with the Rating Scale Model (NPCA-RSM) and structural equation model based on generalized maximum entropy (SEM-GME) or Generalized Cross Entropy (SEM-GCE) could be incorporated into the relevant studies (Ciavolino et al. 2015; Ciavolino and Calcagni 2015; Ciavolino and Carpita 2015). These novel approaches facilitate the reduction of the variables and the evaluation of measurement errors before assessing the different types of HCM estimation with different modes (Mode A and Mode B).

Acknowledgements The authors wish to thank Prof. Dr. Christian M. Ringle and Prof. Dr. Marko Sarstedt for their valuable comments on the earlier version of this manuscript.

Appendix

See Tables 7, 8, 9 and 10.

Table 7 Measurement items*Communicability*

I have heard about people booking travel online many times. (COMM1)

Many friends have purchased travel online. (COMM2)

It is common for people to purchase travel online. (COMM3)

Perceived received advantages (PRA)

Overall, purchasing travel online has perceived relative advantages

Convenience

Purchasing travel online makes me less dependent of opening hours. (CONV1)

Purchasing travel online has easy payment procedures. (CONV2)

Purchasing travel online is more convenient than regular shopping, as I can do it anytime and anywhere. (CONV3)

Time saving

Purchasing travel online enables me to complete shopping quickly. (TS1)

I can save time by purchasing travel online. (TS2)

Purchasing travel online takes less time than purchasing at travel agencies. (TS3)

Financial advantages

I save money by purchasing travel online. (FA1)

Online travel shopping provides more discounts than offline travel purchasing. (FA2)

Generally, travel websites offer tourism products at cheaper prices. (FA3)

Produce variety

There is a larger choice of travel products available when purchasing online. (PV1)

The Internet allows me to purchase travel services that are not available offline. (PV2)

I can design a custom made trip by purchasing travel online. (PV3)

Enjoyment

Purchasing travel online is more exciting than purchasing offline. (ENJOY1)

Purchasing travel online enjoys me more than purchasing offline. (ENJOY2)

Intention to purchase travel online

If you were to purchase travel the probability of purchasing online would be high. (INT1)

I expect to purchase travel online in the near future (INT2)

Table 8 Sample demographics

Demographic variable	Category	Frequency	Percentage (%)
Gender	Male	195	39.0
	Female	305	61.0
Ethnicity	Malay	292	58.4
	Chinese	132	26.4
	Indian	29	5.8
	Others	47	9.4
Age	21–30 years old	292	58.4
	31–40 years old	164	32.8
	41–50 years old	33	6.6
	51–60 years old	11	2.2

Table 8 (continued)

Demographic variable	Category	Frequency	Percentage (%)
Education	Degree	232	46.4
	Master	192	38.4
	Ph.D.	76	15.2
Level of income	RM2001-RM3000	238	47.6
	RM3001-RM4000	84	16.8
	RM4001-RM5000	47	9.4
	RM5001-RM6000	45	9.0
	RM6001-RM7000	30	6.0
	RM7001 and above	56	11.2
How frequent did you purchase travel online?	Yearly	308	61.6
	Half-yearly	103	20.6
	Quarterly	61	12.2
	Monthly	28	5.6
Purpose on purchase travel online	Family-trip	316	63.2
	Business-trip	52	10.4
	Honeymoon-trip	27	5.4
	Friends-trip	105	21.0

Table 9 Assessment of confirmatory tetrad analysis

Construct	Model-implied non-redundant vanishing tetrad	Tetrad value	Ci ^a _{adj}		Supported (reflective)
			LB	UB	
COMM	1: COMM1, COMM2, COMM3, Enjoy1	0.056	-0.017	0.131	Yes
	2: COMM1, COMM2, Enjoy1, COMM3	0.041	-0.044	0.132	
CONV	1: CONV1, CONV2, CONV3, Int1	0.019	-0.057	0.096	Yes
	2: CONV1, CONV2, Int1, CONV3	-0.002	-0.107	0.104	
ENJOY	1: Enjoy1, Enjoy2, Int1, Int2	0.248	-0.165	0.559	Yes
	2: Enjoy1, Enjoy2, Int2, Int1	0.271	-0.033	0.585	
FA	1: FA1, FA2, FA3, Int1	-0.067	-0.160	0.029	Yes
	2: FA1, FA2, Int1, FA3	-0.087	-0.193	0.018	
PV	1: Int1, PV1, PV2, PV3	0.116	-0.194	0.428	Yes
	2: Int1, PV1, PV3, PV2	0.067	-0.018	0.152	
TS	1: Int1, TS1, TS2, TS3	0.055	-0.036	0.146	Yes
	2: Int1, TS1, TS3, TS2	0.047	-0.061	0.152	
IPTO	1: COMM2, COMM3, Int1, Int2	0.194	-0.099	0.488	Yes
	2: COMM2, COMM3, Int2, Int1	0.164	-0.027	0.359	
PRA	1: CONV, Enjoy, FA, PV	-0.036	-0.150	0.078	No
	2: CONV, Enjoy, PV, FA	0.126	0.030	0.227	
	4: CONV, Enjoy, FA, TS	-0.018	-0.081	0.041	
	6: CONV, FA, TS, Enjoy	0.316	0.014	0.533	
	10: CONV, Financial, PV, TS	0.010	-0.055	0.077	

^aCI_{adj} = 90% bias corrected and Bonferroni-adjusted bootstrap confidence interval

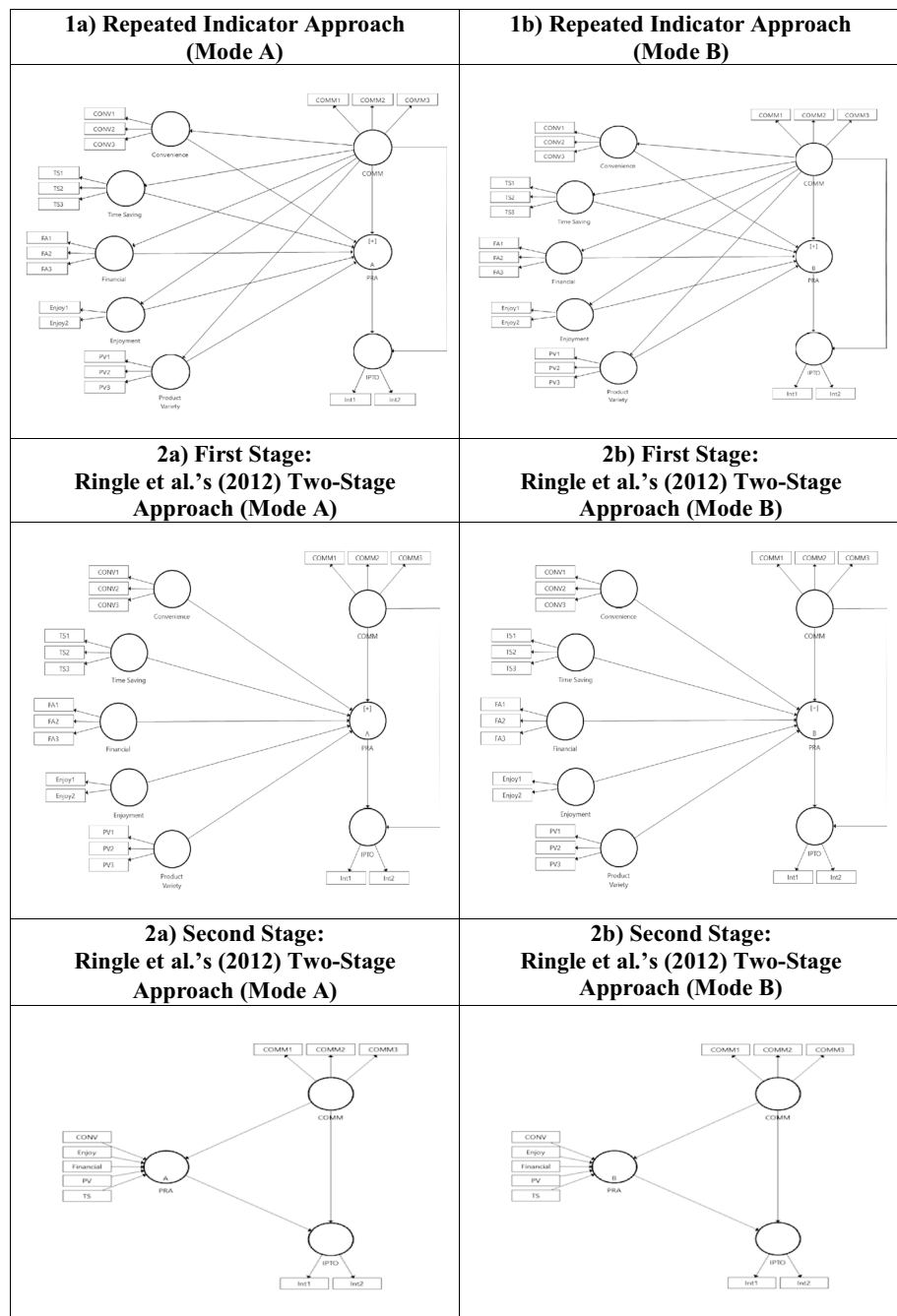
Table 10 Five approaches and two different modes in assessing the HCM

Table 10 (continued)

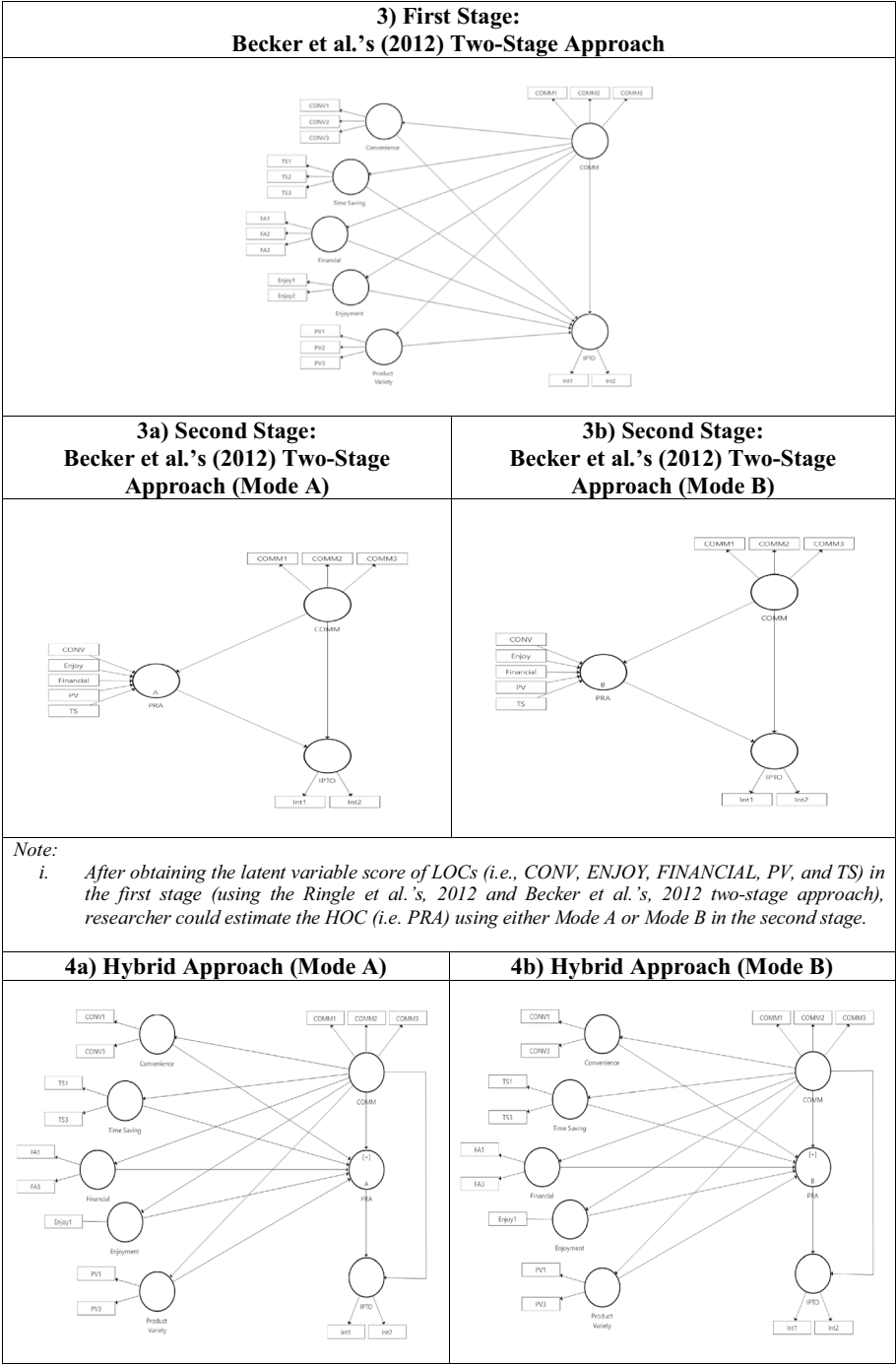
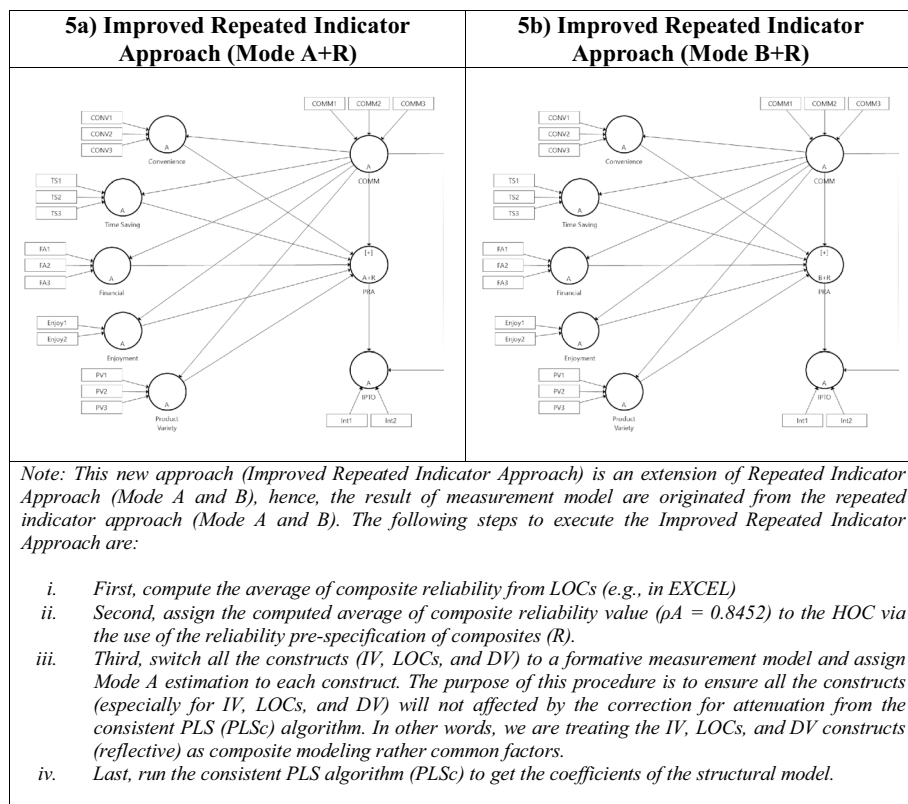


Table 10 (continued)



References

- Aguirre-Urreta, M.I., Rönkkö, M.: Statistical inference with PLSc using bootstrap confidence intervals. *MIS Q.* **42**(3), 1001–1020 (2018)
- Ali, F., Rasoolimanesh, S.M., Sarstedt, M., Ringle, C.M., Ryu, K.: An assessment of the use of partial least squares structural equation modeling (PLS-SEM) in hospitality research. *Int. J. Contemp. Hosp. Manag.* **30**(1), 514–538 (2018)
- Amaro, S., Duarte, P.: An integrative model of consumers' intentions to purchase travel online. *Tour. Manag.* **46**, 64–79 (2015)
- Arora, R.: Validation of an SOR model for situation, enduring, and response components of involvement. *J. Mark. Res.* **19**(4), 505–516 (1982)
- Babin, B.J., Griffin, M., Hair, J.F.: Heresies and sacred cows in scholarly marketing publications. *J. Bus. Res.* **69**(8), 3133–3138 (2016)
- Baron, R.M., Kenny, D.A.: The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J. Pers. Soc. Psychol.* **51**(6), 1173–1182 (1986)
- Barroso, C., Picón, A.: Multi-dimensional analysis of perceived switching costs. *Ind. Mark. Manag.* **41**(3), 531–543 (2012)
- Bauldry, S., Bollen, K.A.: Tetrad: a set of Stata commands for confirmatory tetrad analysis. *Struct. Equ. Model. Multidiscip. J.* **23**(6), 921–930 (2016)
- Becker, J.M., Klein, K., Wetzels, M.: Hierarchical latent variable models in PLS-SEM: guidelines for using reflective–formative type models. *Long Range Plan.* **45**(5–6), 359–394 (2012)

- Becker, J.M., Rai, A., Rigdon, E.: Predictive validity and formative measurement in structural equation modeling: embracing practical relevance. In: Proceedings of the International Conference on Information Systems (ICIS) (2013)
- Bernini, C., Matteucci, M., Mignani, S.: A Bayesian multidimensional IRT approach for the analysis of residents' perceptions toward tourism. *Electron. J. Appl. Stat. Anal.* **8**(3), 272–287 (2015)
- Bigné, E., Sanz, S., Ruiz, C., Aldás, J.: Why some Internet users don't buy air tickets online. *Inf. Commun. Technol. Tourism* **6**, 209–221 (2010)
- Cadogan, J.W., Lee, N.: Improper use of endogenous formative variables. *J. Bus. Res.* **66**(2), 233–241 (2013)
- Cheah, J.H., Sarstedt, M., Ringle, C.M., Ramayah, T., Ting, H.: Convergent validity assessment of formatively measured constructs in PLS-SEM: on using single-item versus multi-item measures in redundancy analyses. *Int. J. Contemp. Hosp. Manag.* (2018). <https://doi.org/10.1108/IJCHM-10-2017-0649>
- Chin, W.W.: How to write up and report PLS analyses. In: Esposito Vinzi, V., Chin, W., Henseler, J., Wang, H. (eds.), *Handbook of Partial Least Squares*, pp. 655–690. Springer, Berlin (2010)
- Ciavolino, E.: General distress as second order latent variable estimated through PLS-PM approach. *Electron. J. Appl. Stat. Anal.* **5**(3), 458–464 (2012)
- Ciavolino, E., Calcagni, A.: Generalized cross entropy method for analysing the SERVQUAL model. *J. Appl. Stat.* **42**(3), 520–534 (2015)
- Ciavolino, E., Carpita, M.: The GME estimator for the regression model with a composite indicator as explanatory variable. *Qual. Quant.* **49**(3), 955–965 (2015)
- Ciavolino, E., Nitti, M.: Using the hybrid two-step estimation approach for the identification of second-order latent variable models. *J. Appl. Stat.* **40**(3), 508–526 (2013)
- Ciavolino, E., Carpita, M., Al-Nasser, A.: Modelling the quality of work in the Italian social co-operatives combining NPCA-RSM and SEM-GME approaches. *J. Appl. Stat.* **42**(1), 161–179 (2015a)
- Ciavolino, E., Carpita, M., Nitti, M.: High-order pls path model with qualitative external information. *Qual. Quant.* **49**(4), 1609–1620 (2015b)
- Ciavolino, E., Salvatore, S., Mossi, P., Lagetto, G.: High-order PLS path model for multi-group analysis: the prosumership service quality model. *Qual. Quant.* (2018). <https://doi.org/10.1007/s11135-018-0747-0>
- Cohen, J.: *Statistical Power Analysis for the Behavioral Sciences*, 2nd edn. Lawrence Erlbaum Associates, Publishers, Mahwah (1988)
- Diamantopoulos, A., Siguaw, J.A.: Formative versus reflective indicators in organizational measure development: a comparison and empirical illustration. *Br. J. Manag.* **17**(4), 263–282 (2006)
- Diamantopoulos, A., Riefler, P., Roth, K.P.: Advancing formative measurement models. *J. Bus. Res.* **61**(12), 1203–1218 (2008)
- do Valle, P.O., Assaker, G.: Using partial least squares structural equation modeling in tourism research: a review of past research and recommendations for future applications. *J. Travel Res.* **55**(6), 695–708 (2016)
- Dolbier, C.L., Webster, J.A., McCalister, K.T., Mallon, M.W., Steinhardt, M.A.: Reliability and validity of a single-item measure of job satisfaction. *Am. J. Health Promot.* **19**(3), 194–198 (2005)
- Edwards, J.R.: Multidimensional constructs in organizational behavior research: an integrative analytical framework. *Organ. Res. Methods* **4**(2), 144–192 (2001)
- Fink, A.: *How to Conduct Surveys: A Step-by-Step Guide*. Sage, Thousand Oaks (2017)
- Franke, G., Sarstedt, M.: Heuristics versus statistics in discriminant validity testing: A comparison of four procedures. *Internet Res.* (forthcoming)
- Fuller, C.M., Simmering, M.J., Atinc, G., Atinc, Y., Babin, B.J.: Common methods variance detection in business research. *J. Bus. Res.* **69**(8), 3192–3198 (2016)
- Garson, G.D.: *Partial Least Squares: Regression and Structural Equation Model*. Statistical Associates Publishing, Asheboro (2016)
- Geisser, S.: A predictive approach to the random effect model. *Biometrika* **61**(1), 101–107 (1974)
- González Rodríguez, M.R., Sámper, R.M.: Analysis of the efficiency of Spanish travel agencies. *Electron. J. Appl. Stat. Anal.* **5**(1), 60–73 (2012)
- Goodhue, D.L., Lewis, W., Thompson, R.: Does PLS have advantages for small sample size or non-normal data? *MIS Q.* **26**(2), 981–1001 (2012)
- Grewal, D., Monroe, K.B., Krishnan, R.: The effects of price-comparison advertising on buyers' perceptions of acquisition value, transaction value, and behavioral intentions. *J. Mark.* **62**(2), 46–59 (1998)
- Gudergan, S.P., Ringle, C.M., Wende, S., Will, A.: Confirmatory tetrad analysis in PLS path modeling. *J. Bus. Res.* **61**(12), 1238–1249 (2008)

- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E.: *Multivariate Data Analysis*. Prentice Hall, Upper Saddle River (2010)
- Hair, J.F., Sarstedt, M., Pieper, T.M., Ringle, C.M.: The use of partial least squares structural equation modeling in strategic management research: a review of past practices and recommendations for future applications. *Long Range Plan.* **45**(5–6), 320–340 (2012a)
- Hair, J.F., Sarstedt, M., Ringle, C.M., Mena, J.A.: An assessment of the use of partial least squares structural equation modeling in marketing research. *J. Acad. Mark. Sci.* **40**(3), 414–433 (2012b)
- Hair, J.F., Sarstedt, M., Ringle, C.M.: Partial least squares structural equation modeling: rigorous applications, better results and higher acceptance. *Long Range Plan.* **46**(1), 1–12 (2013)
- Hair, J., Hollingsworth, C.L., Randolph, A.B., Chong, A.Y.L.: An updated and expanded assessment of PLS-SEM in information systems research. *Ind. Manag. Data Syst.* **117**(3), 442–458 (2017a)
- Hair, J.F., Hult, G.T.M., Ringle, C.M., Sarstedt, M.A.: *Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*. Sage, Thousand Oaks (2017b)
- Hair, J.F., Hult, G.T.M., Ringle, C.M., Sarstedt, M., Thiele, K.O.: Mirror, mirror on the wall: a comparative evaluation of composite-based structural equation modeling methods. *J. Acad. Mark. Sci.* **45**(5), 616–632 (2017c)
- Hair, J.F., Sarstedt, M., Ringle, C.M., Gudergan, S.P.: *Advanced Issues in Partial Least Squares Structural Equation Modeling (PLS-SEM)*. Sage, Thousand Oaks (2018)
- Henseler, J., Ringle, C.M., Sinkovics, R.R.: The use of partial least squares path modeling in international marketing. In: Sinkovics, R.R., Ghauri, P.N. (eds.), *New Challenges to International Marketing*, pp. 277–319. Emerald Group Publishing Limited (2009)
- Henseler, J., Dijkstra, T.K., Sarstedt, M., Ringle, C.M., Diamantopoulos, A., Straub, D.W., et al.: Common beliefs and reality about PLS: comments on Rönkkö and Evermann (2013). *Organ. Res. Methods* **17**(2), 182–209 (2014)
- Henseler, J., Ringle, C.M., Sarstedt, M.: A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J. Acad. Mark. Sci.* **43**(1), 115–135 (2015)
- Higgins, C.A., Compeau, D.R., Meister, D.B.: From prediction to explanation: reconceptualizing and extending the perceived characteristics of innovating. *J. Assoc. Inf. Syst.* **8**(8), 26 (2007)
- Holak, S.L., Lehmann, D.R.: Purchase intentions and the dimensions of innovation: an exploratory model. *J. Prod. Innov. Manag.* **7**(1), 59–73 (1990)
- Ingusci, E., Manuti, A., Callea, A.: Employability as mediator in the relationship between the meaning of working and job search behaviours during unemployment. *Electron. J. Appl. Stat. Anal.* **9**(1), 1–16 (2016)
- Jarvis, C.B., MacKenzie, S.B., Podsakoff, P.M.: A critical review of construct indicators and measurement model misspecification in marketing and consumer research. *J. Consum. Res.* **30**(2), 199–218 (2003)
- Johnson, R.E., Rosen, C.C., Djurdjevic, E., Taing, M.U.: Recommendations for improving the construct clarity of higher-order multidimensional constructs. *Hum. Resour. Manag. Rev.* **22**(2), 62–72 (2012)
- Kaufmann, L., Gaeckler, J.: A structured review of partial least squares in supply chain management research. *J. Purch. Supply Manag.* **21**(4), 259–272 (2015)
- Kline, R.B.: *Principles and Practice of Structural Equation Modeling*. Guilford Press, New York (2016)
- Lee, N., Cadogan, J.W.: Problems with formative and higher-order reflective variables. *J. Bus. Res.* **66**(2), 242–247 (2013)
- Lee, L., Petter, S., Fayard, D., Robinson, S.: On the use of partial least squares path modeling in accounting research. *Int. J. Account. Inf. Syst.* **12**(4), 305–328 (2011)
- Li, L., Buhalis, D.: E-commerce in China: the case of travel. *Int. J. Inf. Manag.* **26**(2), 153–166 (2006)
- Limayem, M., Khalifa, M., Frini, A.: What makes consumers buy from Internet? A longitudinal study of online shopping. *IEEE Trans. Syst. Man Cybern. A Syst. Hum.* **30**(4), 421–432 (2000)
- Lohmöller, J.B.: Latent variable path analysis with partial least squares. *Physica, Heidelberg* (1989)
- Marcoulides, G.A., Chin, W.W., Saunders, C.: When imprecise statistical statements become problematic: a response to Goodhue, Lewis, and Thompson. *MIS Q.* **36**(3), 717–728 (2012)
- Mehrabian, A., Russell, J.A.: *An Approach to Environment Psychology*. MIT Press, Cambridge (1974)
- Morrión, A.M., Jing, S., O’Leary, J.T., Cai, L.A.: Predicting usage of the internet for travel bookings: an exploratory study. *Inf. Technol. Tour.* **4**(1), 15–30 (2001)
- Nitzl, C.: The use of partial least squares structural equation modelling (PLS-SEM) in management accounting research: directions for future theory development. *J. Account. Lit.* **37**, 19–35 (2016)
- Nitti, M., Ciavolino, E.: A deflated indicators approach for estimating second-order reflective models through PLS-PM: an empirical illustration. *J. Appl. Stat.* **41**(10), 2222–2239 (2014)
- Nitzl, C., Roldan, J.L., Cepeda, G.: Mediation analysis in partial least squares path modeling: helping researchers discuss more sophisticated models. *Ind. Manag. Data Syst.* **116**(9), 1849–1864 (2016)

- Peng, D.X., Lai, F.: Using partial least squares in operations management research: a practical guideline and summary of past research. *J. Oper. Manag.* **30**(6), 467–480 (2012)
- Podsakoff, P.M., Organ, D.W.: Self-reports in organizational research: problems and prospects. *J. Manag.* **12**(4), 531–544 (1986)
- Polites, G.L., Roberts, N., Thatcher, J.: Conceptualizing models using multidimensional constructs: a review and guidelines for their use. *Eur. J. Inf. Syst.* **21**(1), 22–48 (2012)
- Richter, N.F., Sinkovics, R.R., Ringle, C.M., Schlaegel, C.: A critical look at the use of SEM in international business research. *Int. Mark. Rev.* **33**(3), 376–404 (2016a)
- Richter, N.F., Cepeda, G., Roldán, J.L., Ringle, C.M.: European management research using partial least squares structural equation modeling (PLS-SEM). *Eur. Manag. J.* **34**(6), 589–597 (2016b)
- Rigdon, E.E.: Rethinking partial least squares path modeling: in praise of simple methods. *Long Range Plan.* **45**(5–6), 341–358 (2012)
- Rigdon, E.E.: Comment on “Improper use of endogenous formative variables”. *J. Bus. Res.* **67**(1), 2800–2802 (2014a)
- Rigdon, E.E.: Rethinking partial least squares path modeling: breaking chains and forging ahead. *Long Range Plan.* **47**(3), 161–167 (2014b)
- Rigdon, E.E.: Choosing PLS path modeling as analytical method in European management research: a realist perspective. *Eur. Manag. J.* **34**(6), 598–605 (2016)
- Ringle, C.M., Sarstedt, M., Straub, D.W.: A critical look at the use of PLS-SEM in MIS quarterly. *MIS Q.* **36**(1), iii–xiv (2012)
- Ringle, C.M., Wende, S., Becker, J.M.: SmartPLS 3. SmartPLS GmbH, Boenningstedt (2015)
- Ringle, C.M., Sarstedt, M., Mitchell, R., Gudergan, S.P.: Partial least squares structural equation modeling in HRM research. *Int. J. Hum. Resour. Manag.* (2018). <https://doi.org/10.1080/09585192.2017.1416655>
- Rönkkö, M., Evermann, J.: A critical examination of common beliefs about partial least squares path modeling. *Organ. Res. Methods* **16**(3), 425–448 (2013)
- Sarstedt, M., Ringle, C.M., Smith, D., Reams, R., Hair Jr., J.F.: Partial least squares structural equation modeling (PLS-SEM): a useful tool for family business researchers. *J. Fam. Bus. Strategy* **5**(1), 105–115 (2014)
- Sarstedt, M., Hair, J.F., Ringle, C.M., Thiele, K.O., Gudergan, S.P.: Estimation issues with PLS and CBSEM: where the bias lies! *J. Bus. Res.* **69**(10), 3998–4010 (2016)
- Sarstedt, M., Ringle, C.M., Hair, J.F.: Partial least squares structural equation modeling. In: Homburg, C., Klarmann, M., Vomberg, A. (eds.), *Handbook of Market Research*, pp. 1–40. Springer, Cham (2017)
- Schönemann, P.H., Wang, M.M.: Some new results on factor indeterminacy. *Psychometrika* **37**(1), 61–91 (1972)
- Shmueli, G., Ray, S., Velasquez Estrada, J.M., Chatla, S.: The elephant in the room: evaluating the predictive performance of PLS models. *J. Bus. Res.* **69**(10), 4552–4564 (2016)
- Slama, M.E., Tashchian, A.: Validating the SOR paradigm for consumer involvement with a convenience good. *J. Acad. Mark. Sci.* **15**(1), 36–45 (1987)
- Sosik, J.J., Kahai, S.S., Piovoso, M.J.: Silver bullet or voodoo statistics? A primer for using the partial least squares data analytic technique in group and organization research. *Group Organ. Manag.* **34**(1), 5–36 (2009)
- Stone, M.: Cross-validatory choice and assessment of statistical predictions. *J. R. Stat. Soc. B (Methodological)*, **36**(2), 111–147 (1974)
- Streukens, S., Leroy-Werelds, S.: Bootstrapping and PLS-SEM: a step-by-step guide to get more out of your bootstrap results. *Eur. Manag. J.* **34**(6), 618–632 (2016)
- Teo, T.S., Yeong, Y.D.: Assessing the consumer decision process in the digital marketplace. *Omega* **31**(5), 349–363 (2003)
- Van Riel, A.C., Henseler, J., Kemény, I., Sasovova, Z.: Estimating hierarchical constructs using consistent partial least squares: the case of second-order composites of common factors. *Ind. Manag. Data Syst.* **117**(3), 459–477 (2017)
- Voorhees, C.M., Brady, M.K., Calantone, R., Ramirez, E.: Discriminant validity testing in marketing: an analysis, causes for concern, and proposed remedies. *J. Acad. Mark. Sci.* **44**(1), 119–134 (2016)
- Wanous, J.P., Reichers, A.E.: Estimating the reliability of a single-item measure. *Psychol. Rep.* **78**(2), 631–634 (1996)
- Wetzels, M., Odekerken-Schröder, G., Van Oppen, C.: Using PLS path modeling for assessing hierarchical construct models: guidelines and empirical illustration. *MIS Q.* **33**(1), 177–195 (2009)
- Willaby, H.W., Costa, D.S., Burns, B.D., MacCann, C., Roberts, R.D.: Testing complex models with small sample sizes: a historical overview and empirical demonstration of what partial least squares (PLS) can offer differential psychology. *Personal. Individ. Differ.* **84**, 73–78 (2015)

- Wilson, B.: Using PLS to investigate interaction effects between higher order branding constructs. In: Esposito Vinzi, V., Chin, W., Henseler, J., Wang, H. (eds.), *Handbook of Partial Least Squares*, pp. 621–652. Springer, Berlin (2010)
- Wilson, B., Henseler, J.: Modeling reflective higher-order constructs using three approaches with PLS path modeling: a Monte Carlo comparison. In: Thyne, M., Deans, K.R., Gnoth, J. (eds.) *Australian and New Zealand Marketing Academy Conference Proceedings*, pp. 791–800. Department of Marketing, School of Business, University of Otago, Otago (2007)
- Wold, H.O.: Soft modelling: the basic design and some extensions. In: Jöreskog, K.G., Wold, H.O. (eds.), *Systems under indirect observations: Part II*. North-Holland, Amsterdam, pp. 1–54 (1982)
- Wold, H.O.: Partial Least Squares. In: Kotz, S., Read, C.B., Balakrishnan, N., Vidakovic, B., Johnson, N.L. (eds.), *Encyclopedia of Statistical Sciences*, Wiley (2006)
- Zhao, X., Lynch Jr., J.G., Chen, Q.: Reconsidering Baron and Kenny: myths and truths about mediation analysis. *J. Consum. Res.* **37**(2), 197–206 (2010)

Affiliations

Jun-Hwa Cheah¹ · Hiram Ting² · T. Ramayah³ · Mumtaz Ali Memon⁴ · Tat-Huei Cham⁵ · Enrico Ciavolino⁶ 

Jun-Hwa Cheah
jackycheahjh@gmail.com

Hiram Ting
hiramparousia@gmail.com

T. Ramayah
ramayah@usm.my

Mumtaz Ali Memon
mumtazutp@gmail.com

Tat-Huei Cham
jaysoncham@gmail.com

¹ Faculty of Economics and Management, Universiti Putra Malaysia (UPM), Serdang, Selangor, Malaysia

² Faculty of Hospitality and Tourism Management, UCSI University, Kuala Lumpur, Malaysia

³ School of Management, Universiti Sains Malaysia (USM), Penang, Malaysia

⁴ Air University School of Management, Air University, Islamabad, Pakistan

⁵ Faculty of Accountancy and Management, Universiti Tunku Abdul Rahman (UTAR), Selangor, Malaysia

⁶ University of Salento, Lecce, Italy